OAVS · TGT COMPUTER TEACHER EXAMINATION

Normalisation

Comprehensive MCQ Question Bank

SYLLABUS CHAPTER 7 · TOPIC 7.1

Advantages & Disadvantages of Normalisation
1NF – 2NF – 3NF Rules with Examples · Anomalies

Strictly limited to the prescribed syllabus boundary

249	12	52
MCQS	SECTIONS	TABLE QUESTIONS

Question Bank · With Answers & Explanations · Print-Friendly Format

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	Total Questions	**249 MCQs**

Scope: every question stays within Advantages/Disadvantages of Normalisation, 1NF, 2NF, 3NF, their rules with examples, and the insertion, update and deletion anomalies. Correct answers are spread evenly across A–D with no repeating pattern. Difficulty mix: 65 Easy, 122 Moderate, 62 Difficult/Tricky. 52 questions include a rendered relation (table).

SECTION 01

Introduction to Normalisation

7 Questions · 4 Easy · 3 Moderate

Q1 · *Concept | Normalisation* EASY

What is normalization primarily used for?

A. Removing the primary key from every table

B. Combining all attributes into one large table

C. Reducing redundancy and improving data integrity

D. Increasing the number of repeated records across tables

Ans: C

EXPLANATION Normalization organizes relational data to minimize redundancy and improve consistency and integrity.

Q2 · *Concept | Normalisation* EASY

Normalization is applied to which of the following?

A. Records stored in a flat unstructured file

B. Tables in a relational database

C. Queries written to retrieve stored data

D. Diagrams drawn to plan a database

Ans: B

EXPLANATION Normalization is a relational database design technique applied to tables/relations.

Q3 · *Concept | Normalisation* EASY

The primary goal of normalization is to reduce:

A. Number of users accessing the database

B. Data redundancy and update anomalies

C. Number of primary keys in a table

D. Physical size of the storage disk

Ans: B

EXPLANATION Normalization mainly targets reducing redundancy, which in turn reduces the anomalies it causes.

Q4 · *Concept | Normalisation* EASY

Normalization organizes data by:

A. Merging all related tables into one large table to avoid joins

B. Removing primary keys so that rows can repeat freely

C. Storing all the data in unstructured text files

D. Splitting large tables into smaller related tables

Ans: D

EXPLANATION Normalization decomposes tables to remove redundancy while preserving relationships via keys.

Q5 · *Definition | Normalisation* MODERATE

A 'normal form' refers to:

A. A format in which duplicated records are stored together

B. A level of table organization defined by the rules it satisfies

C. A table that contains all attributes combined in a single relation

D. A fixed template used to design the input forms of an application

Ans: B

EXPLANATION A normal form is a set of conditions a relation must satisfy for a given normalization level.

Q6 · *Concept | Normalisation* MODERATE

Normalization primarily works at which stage of database design?

A. Deciding which attributes are shown on a form

B. Assigning a primary key to a single big table

C. Logical design of relations

D. Physical arrangement of data on disk blocks

Ans: C

EXPLANATION Normalization is a logical design activity concerned with how attributes are grouped into relations.

Q7 · *Concept | Normalisation* MODERATE

Moving from 1NF to 2NF to 3NF generally means:

A. Applying stricter rules that reduce redundancy further

B. Removing all constraints so that tables become flat files

C. Applying looser rules that gradually increase redundancy

D. Combining the attributes of all tables into a single relation

Ans: A

EXPLANATION Each higher normal form imposes additional constraints beyond the previous one, further reducing redundancy.

SECTION 02

Advantages of Normalisation

20 Questions · 10 Easy · 9 Moderate · 1 Difficult

Q8 · *Concept | Normalisation* EASY

Which is a key advantage of normalization?

A. Increase in number of anomalies
B. Reduction of data redundancy
C. Increase in duplicate records
D. Reduction in data accuracy

Ans: B

EXPLANATION Normalization primarily aims to reduce redundant storage of the same data.

Q9 · *Concept | Normalisation* EASY

Normalization improves:

A. Data consistency across the database
B. The chance of conflicting data values
C. The number of duplicate copies of data
D. The complexity of retrieving a single record

Ans: A

EXPLANATION By storing facts once, normalization keeps data consistent throughout the database.

Q10 · *Concept | Normalisation* EASY

One advantage of normalization is better:

A. Keeping the same data repeated in several tables
B. Removing the primary key from related tables
C. Organization of data into logical groups
D. Storing all attributes together in a single large table

Ans: C

EXPLANATION Normalization groups related attributes into well-structured relations.

Q11 · *Concept | Normalisation* MODERATE

Normalization helps reduce:

A. Speed of data retrieval in all cases
B. Duplication of data across tables
C. Number of relations in the database
D. Number of primary keys used

Ans: B

EXPLANATION A core benefit of normalization is minimizing duplicate storage of the same facts.

Q12 · *Concept | Normalisation* EASY

Applying normalization improves:

A. Update inconsistency
B. Data redundancy
C. Data integrity
D. Storage duplication

Ans: C

EXPLANATION With reduced redundancy, data integrity (accuracy and reliability) improves.

Q13 · *Concept | Normalisation* MODERATE

A major benefit of normalization in database design is:

A. Making the database structure harder to maintain
B. Removing the need for primary keys
C. Making the database structure easier to maintain
D. Eliminating the need for relationships between tables

Ans: C

EXPLANATION Well-normalized structures are easier to maintain because updates need fewer changes.

Q14 · *Concept | Normalisation* EASY

Normalization benefits an organization by:

A. Reducing unnecessary duplicate data
B. Increasing the number of repeated records
C. Removing the primary key from every table
D. Combining all attributes into one table

Ans: A

EXPLANATION Less redundancy means less duplicate data is stored.

Q15 · *Application | Normalisation* MODERATE

If a customer's address is stored once in a Customers table instead of repeating it in every order record, this reflects the normalization advantage of:

A. Increased anomalies
B. Reduced data accuracy
C. Increased query complexity
D. Reduced data redundancy

Ans: D

EXPLANATION Storing the address once and referencing it avoids repeating the same data in every order.

Q16 · *Concept | Normalisation* MODERATE

Normalization reduces the likelihood of:

A. Inconsistent copies of the same data
B. Consistent copies of the same data across tables
C. Use of primary keys to identify each row
D. Use of foreign keys to link related tables

Ans: A

EXPLANATION Fewer duplicate copies mean less chance that different copies become inconsistent.

Q17 · *Concept | Normalisation* EASY

One advantage of normalization is easier:

A. Updating data, since it is copied in many different tables

B. Updating data, since related records are merged into one row

C. Updating data, since every table has no primary key

D. Updating data, since it is stored in only one place

Ans: D

EXPLANATION Because each fact is stored once, updates need to be made in only one place.

Q18 · *Concept | Normalisation* MODERATE

Normalization supports better database design by:

A. Combining all attributes into one large table

B. Grouping attributes by functional dependencies

C. Ignoring the relationships between attributes

D. Removing every functional dependency from tables

Ans: B

EXPLANATION Normalization groups attributes according to their dependency on keys.

Q19 · *Concept | Normalisation* EASY

A normalized database generally has:

A. Exactly the same redundancy as an unnormalized one

B. More redundant data than an unnormalized one, since tables are split

C. Less redundant data than an unnormalized one

D. No tables at all, because all data is kept in one file

Ans: C

EXPLANATION The purpose of normalization is to reduce redundant data compared to an unnormalized design.

Q20 · *Concept | Normalisation* MODERATE

Normalization helps prevent which of the following from occurring frequently?

A. Splitting of data into several related tables

B. Increase in the number of repeated records

C. Insertion, update and deletion anomalies

D. Use of primary keys to identify each row

Ans: C

EXPLANATION Reduced redundancy directly reduces the occurrence of anomalies.

Q21 · *Concept | Normalisation* EASY

Which of these is NOT typically considered an advantage of normalization?

A. Better logical organization of attributes

B. Increased data redundancy

C. Reduced duplication of stored facts

D. Improved consistency of stored data

Ans: B

EXPLANATION Increased redundancy is a drawback normalization avoids, not a benefit it provides.

Q22 · *Concept | Normalisation* MODERATE

Normalization improves data quality mainly by:

A. Ensuring each fact is stored in exactly one place

B. Removing all data validation rules

C. Ensuring each fact is stored in multiple places

D. Ignoring functional dependencies between attributes

Ans: A

EXPLANATION Storing each fact once removes the possibility of it disagreeing with another copy.

Q23 · *Concept | Normalisation* EASY

Normalized tables are generally easier to:

A. Query because they contain no relationships

B. Delete without any side effects

C. Modify without affecting unrelated data

D. Modify because they contain more duplicate data

Ans: C

EXPLANATION Because related facts are isolated in appropriate tables, modifications affect only the relevant table.

Q24 · *Concept | Normalisation* MODERATE

A well-normalized design tends to improve:

A. The number of duplicate rows stored in tables

B. Long-term data reliability and accuracy

C. The number of update anomalies in the database

D. Short-term redundancy of records in every table

Ans: B

EXPLANATION By minimizing redundancy, normalization improves long-term reliability of stored data.

Q25 · *Concept | Normalisation* EASY

Which statement best describes an advantage of normalization?

A. It removes the need for primary keys in the tables

B. It increases the chances of storing the same fact in many places

C. It combines all the attributes into one large relation

D. It reduces the chances of storing the same fact repeatedly

Ans: D

EXPLANATION Minimizing duplicate storage of facts is a defining advantage of normalization.

Q26 · *Application | Normalisation* MODERATE

A shop stores a customer's address in all 500 of her order rows. After normalization the address is kept once in a Customer table. Which benefit does this show most directly?

A. More joins, so the data stored becomes more accurate

B. More tables, so every query always runs faster

C. Removal of the need for a primary key in tables

D. Less redundancy, so an address change is made once

Ans: D

EXPLANATION Storing a fact once removes duplication and makes consistent updating easy.

Q27 · *Statement-based | Normalisation*

DIFFICULT

Statement I: Normalization reduces data redundancy. Statement II: Normalization improves data consistency by storing each fact in one place. Which is correct?

A. Both are correct; II explains how consistency improves

B. Only Statement I is correct, since normalization has no effect on consistency

C. Neither statement is correct, since normalization increases redundancy

D. Only Statement II is correct, since normalization does not reduce redundancy

Ans: A

EXPLANATION Less duplication means copies cannot disagree, which improves consistency and integrity.

SECTION 03

Disadvantages of Normalisation

16 Questions · 7 Easy · 8 Moderate · 1 Difficult

Q28 · *Concept | Normalisation* EASY

A common disadvantage of normalization is:

A. More repeated records stored across the tables

B. Less consistent data because the same fact is stored in many rows

C. Fewer tables in the database, with all attributes combined

D. More tables in the database

Ans: D

EXPLANATION Normalization typically splits data into more, smaller tables.

Q29 · *Concept | Normalisation* MODERATE

Normalization can increase:

A. Number of update anomalies in the tables

B. Complexity of the database design

C. Chances of duplicate data being stored

D. Redundancy among the stored attributes

Ans: B

EXPLANATION More tables and relationships increase the structural complexity of the design.

Q30 · *Concept | Normalisation* EASY

A drawback of normalization is the need for additional:

A. Update anomalies in every table after splitting it

B. Joins when retrieving data from multiple tables

C. Duplicate rows in a single table to speed up retrieval

D. Redundant copies of the same attribute in many tables

Ans: B

EXPLANATION Since related data is split across tables, queries often need joins to bring it back together.

Q31 · *Concept | Normalisation* MODERATE

Excessive joins caused by normalization can affect:

A. Query performance

B. Table naming conventions

C. Data duplication

D. Data redundancy

Ans: A

EXPLANATION More joins generally mean more processing, which can affect performance for complex queries.

Q32 · *Application | Normalisation* MODERATE

A report that must combine data from five normalized tables will likely require:

A. Fewer joins than an unnormalized design would need
B. No joins at all, since the data is already in one place
C. Automatic merging of the tables back into a single table
D. Several joins, making queries more complex

Ans: D

EXPLANATION Retrieving combined data from several normalized tables typically requires multiple joins.

Q33 · *Concept | Normalisation* EASY

Which is a valid disadvantage of a highly normalized database?

A. Complex queries needing joins for combined reports
B. A smaller number of tables in the design
C. Lower data integrity in the relations
D. Increased redundancy of stored data in many tables

Ans: A

EXPLANATION Highly normalized designs often need more complex queries to reconstruct combined views.

Q34 · *Concept | Normalisation* MODERATE

One trade-off of normalization is that it can make database design:

A. More time-consuming, as more relations must be planned
B. Free of any relationships between the tables of the database
C. Instant, since no planning of tables is required at all
D. Impossible to implement for any realistic application

Ans: A

EXPLANATION Designing many related, smaller tables requires more careful planning.

Q35 · *Concept | Normalisation* EASY

Beginners often find normalization challenging because it:

A. Needs no understanding of how attributes relate to each other
B. Requires removing the primary key from each table
C. Needs understanding of dependencies among attributes
D. Requires storing every attribute in a single large table

Ans: C

EXPLANATION Correctly normalizing requires analyzing functional dependencies, which can be conceptually challenging.

Q36 · *Concept | Normalisation* MODERATE

A disadvantage often cited for normalization is:

A. The need to manage fewer foreign key relationships
B. The elimination of all relationships
C. The need to manage more foreign key relationships
D. The elimination of all keys

Ans: C

EXPLANATION Splitting data into more tables usually increases the number of foreign keys linking them.

Q37 · *Concept | Normalisation* EASY

Which of these best describes a disadvantage of normalization?

A. It always removes the need for primary keys
B. It can increase the number of tables
C. It always reduces the number of tables to one
D. It always makes every kind of query faster

Ans: B

EXPLANATION Splitting a single unnormalized table into related tables increases the total number of tables.

Q38 · *Concept | Normalisation* MODERATE

For read-heavy systems, one disadvantage of a highly normalized design is:

A. Additional joins may slow down complex read queries
B. Additional joins always speed up complex read queries
C. It removes the need for read queries in the database
D. It guarantees zero query overhead for every operation

Ans: A

EXPLANATION More joins generally add overhead, which can be a concern for read-heavy systems.

Q39 · *Concept | Normalisation* EASY

Normalization can make certain queries:

A. Automatically optimized with no cost
B. More complex to write
C. Impossible to write
D. Unnecessary to write

Ans: B

EXPLANATION Because data is split across tables, some queries need more complex joins.

Q40 · *Concept | Normalisation* MODERATE

A basic disadvantage directly associated with normalization is:

A. A more complex overall schema with many tables
B. Guaranteed repetition of the same data in every table
C. A simpler overall schema with only a single large table
D. Complete removal of relationships between the tables

Ans: A

EXPLANATION More relations and relationships make the schema structurally more complex, even though it reduces redundancy.

Q41 · *Concept | Normalisation* EASY

Which of the following is a valid trade-off consideration when normalizing a database?

A. Increased anomalies versus reduced joins
B. Increased redundancy versus reduced number of tables
C. Reduced integrity versus reduced complexity
D. Reduced redundancy versus increased number of joins

Ans: D

EXPLANATION Normalization reduces redundancy but this generally comes at the cost of needing more joins.

Q42 · *Application | Normalisation* MODERATE

A report needs the customer name, the order date and the product name, stored in three normalized tables linked by keys. At least how many joins does it need?

A. 0

B. 1

C. 2

D. 3

Ans: C

EXPLANATION Combining three tables takes at least two joins; this is the extra-join cost of normalization.

Q43 · *Scenario-based | Normalisation* DIFFICULT

A designer splits one table with 8 attributes into 4 smaller tables. Which pair of outcomes is correct?

A. Redundancy increases, and fewer joins are needed to read data

B. Redundancy stays the same, but more tables exist

C. Redundancy falls, but joins may be needed to rebuild the view

D. Redundancy falls, and joins are never needed again

Ans: C

EXPLANATION Normalization trades lower redundancy for more tables and joins.

SECTION 04

First Normal Form (1NF)

23 Questions · 9 Easy · 12 Moderate · 2 Difficult

Q44 · *Definition | Normalisation* EASY

Which normal form requires all attribute values in a relation to be atomic?

A. 1NF
B. 2NF
C. 3NF
D. None of these

Ans: A

EXPLANATION 1NF requires each attribute to contain atomic values and eliminates repeating groups.

Q45 · *Definition | Normalisation* EASY

A relation is in 1NF if it has:

A. No functional dependencies at all among attributes
B. No primary key and only composite attributes in each row
C. No repeating groups and only atomic values
D. Only composite keys and no foreign keys at all

Ans: C

EXPLANATION 1NF eliminates repeating groups and requires atomic values in each cell.

Q46 · *Concept | Normalisation* MODERATE

A 'repeating group' in a table refers to:

A. A primary key that is repeated as a foreign key in another table
B. One value that is repeated again and again across many rows
C. Several values of one attribute stored in a single row
D. A whole table that is copied twice inside the database

Ans: C

EXPLANATION A repeating group means multiple values for one attribute stored in a single record.

Q47 · *Concept | Normalisation* MODERATE

Which of the following violates 1NF?

A. Several phone numbers stored in one Phone cell
B. One email address stored for every row
C. One phone number stored in the Phone column of a row
D. A unique CustomerID stored in each row

Ans: A

EXPLANATION Storing multiple values in one attribute (multivalued) violates the atomicity rule of 1NF.

Q48 · *Table-based | Normalisation* MODERATE

Why does the following relation violate 1NF?

STUDENT

StudentID	StudentName	Courses
101	Asha	Math, Physics
102	Ravi	Chemistry
103	Meena	Math, Biology, English

A. The relation has no foreign key linking it to another table

B. StudentName depends on Courses instead of StudentID

C. StudentID values are repeated in more than one row

D. Courses holds more than one value in some rows

Ans: D

EXPLANATION 1NF needs atomic values; 'Math, Physics' and 'Math, Biology, English' are multiple values in one cell.

Q49 · *Concept | Normalisation* EASY

1NF mainly eliminates:

A. Transitive dependencies among non-key attributes

B. Foreign key references between different tables

C. Partial dependencies on a composite primary key

D. Repeating groups and multivalued attributes

Ans: D

EXPLANATION 1NF specifically targets removing repeating groups/multivalued attributes; partial and transitive dependencies are handled in 2NF/3NF.

Q50 · *Concept | Normalisation* MODERATE

A table where a single cell contains a comma-separated list of values is:

A. In 1NF

B. Not in 1NF

C. In 2NF but not 1NF

D. In 3NF but not 1NF

Ans: B

EXPLANATION A cell holding multiple values is non-atomic, so the relation fails to satisfy 1NF.

Q51 · *Concept | Normalisation* EASY

Which statement about 1NF is TRUE?

A. Every attribute in a 1NF relation must hold a single, atomic value

B. 1NF requires the removal of partial dependency

C. 1NF requires no transitive dependency

D. 1NF requires that all attributes depend only on the full primary key

Ans: A

EXPLANATION The requirement of atomic values defines 1NF; partial/transitive dependency rules belong to 2NF/3NF.

Q52 · *Table-based | Normalisation* MODERATE

The design below fails 1NF because:

ORDERS

OrderID	Product1	Product2	Product3
O1	Pen	Book	NULL
O2	Pen	NULL	NULL

A. There is no violation, as each cell is single-valued

B. NULL values make the relation violate 2NF only

C. OrderID is not an atomic value in each row

D. Product1, Product2, Product3 form a repeating group

Ans: D

EXPLANATION Repeating the same kind of attribute across several columns is a repeating group, which 1NF does not allow.

Q53 · *Concept | Normalisation* MODERATE

Which of the following is an example of a multivalued attribute?

A. A single 'DateOfJoining' field storing one date for one employee

B. A single 'Salary' field storing one amount for one employee

C. A single 'EmployeeID' field storing one ID for one employee

D. A 'Skills' field listing several skills for one employee

Ans: D

EXPLANATION A field holding several skill values for one employee is multivalued and violates 1NF.

Q54 · *Table-based | Normalisation* MODERATE

Which change makes this relation satisfy 1NF?

EMPLOYEE

EmpID	EmpName	Phone
E1	Kiran	98760, 98761
E2	Sunita	91230

A. Add a Phone2 column to hold the second phone number of an employee

B. Delete the EmpName column from the relation to avoid repetition

C. Put each phone number in its own row, e.g. (E1, 98760), (E1, 98761)

D. Join the numbers with a slash inside the same Phone cell of the row

Ans: C

EXPLANATION One value per cell is needed; a Phone2 column would only create a repeating group.

Q55 · *Table-based | Normalisation* EASY

Does this relation satisfy 1NF?

BOOK

BookID	Title	Author	Price
B1	DBMS Basics	Korth	450
B2	Data Structures	Tanenbaum	520
B3	Operating Systems	Galvin	610

A. No, a relation needs at least five columns for 1NF

B. Yes, all values are atomic with no repeating groups

C. No, a relation in 1NF must have a composite key

D. No, the Author column must be split into several columns

Ans: B

EXPLANATION 1NF only checks atomic values and repeating groups; this relation passes both.

Q56 · *Concept | Normalisation* MODERATE

Which technique is used to eliminate repeating groups when converting to 1NF?

A. Adding more repeating columns for the extra values

B. Deleting the repeating attribute entirely with no replacement

C. Merging all repeating values into one long comma-separated string

D. Giving each repeated value its own row or a separate table

Ans: D

EXPLANATION Repeating groups are removed by giving each repeated value its own row (or a linked table), not by merging or deleting them.

Q57 · *Concept | Normalisation* EASY

Which of these violates the 'atomic value' rule of 1NF?

A. A single Address field split into City, State, and Pincode columns

B. A single Name field with just the first name

C. A single Address field containing 'City, State, Pincode' combined

D. A single Age field with a numeric value

Ans: C

EXPLANATION Combining multiple pieces of information into one field breaks atomicity.

Q58 · *Concept | Normalisation* MODERATE

Identifying whether a relation satisfies 1NF mainly involves checking for:

A. Partial dependency of a non-key attribute on a composite key

B. Whether a foreign key refers to the primary key of another table

C. Transitive dependency between two non-key attributes of the table

D. Non-atomic values or repeating groups in a column

Ans: D

EXPLANATION 1NF checks are only about atomicity and repeating groups, not the dependency-based rules of 2NF/3NF.

Q59 · *Concept | Normalisation* EASY

Which of the following tables is already in 1NF?

A. A table where one row represents multiple students at once

B. A table where one column stores a list of phone numbers

C. A table where every column holds exactly one value per row

D. A table where one column stores multiple email addresses per row

Ans: C

EXPLANATION Only a table with single, atomic values per cell satisfies 1NF.

Q60 · *Concept | Normalisation* MODERATE

In an unnormalized table with repeating columns like Course1, Course2, Course3, the underlying issue is:

A. A table name that is too long to use

B. A missing primary key in the table, only

C. Absence of a foreign key in the table

D. A repeating group spread over similar columns

Ans: D

EXPLANATION Course1/Course2/Course3 is a repeating group problem disguised as separate columns, still violating 1NF.

Q61 · *Concept | Normalisation* EASY

The main purpose of converting a table to 1NF is to ensure:

A. Data is stored with maximum redundancy in every single table

B. Data is stored without any keys in the relation

C. Data is stored in nested tables within other tables

D. Data is stored in a simple, atomic, tabular structure

Ans: D

EXPLANATION 1NF's purpose is to enforce a simple flat structure with atomic values, avoiding nested/repeating data.

Q62 · *Table-based | Normalisation* DIFFICULT

How many rows will the relation have after it is converted to 1NF with one (ProjectID, EmployeeID) pair per row?

PROJECT

ProjectID	EmployeeIDs
P1	E01, E02, E03
P2	E04

A. 2

B. 3

C. 4

D. 5

Ans: C

EXPLANATION P1 has three employees and P2 has one, so 3 + 1 = 4 rows.

Q63 · *Concept | Normalisation*

MODERATE

Which of the following statements about 1NF is FALSE?

A. 1NF requires a primary key so that each row is identified uniquely

B. 1NF requires that repeating groups are eliminated from the relation

C. 1NF requires that every attribute value in a cell is atomic

D. 1NF requires that no non-key attribute depends on part of a composite key

Ans: D

EXPLANATION The rule about not depending on part of a composite key is the 2NF rule, not 1NF, making this statement false about 1NF.

Q64 · *Table-based | Normalisation*

DIFFICULT

The Orders relation below is to be converted to 1NF with one (OrderID, Item) row per item actually ordered. How many rows result?

ORDERS

OrderID	Item1	Item2	Item3
O1	Pen	Book	NULL
O2	Pen	NULL	NULL
O3	Ink	Pen	Book

A. 3

B. 6

C. 7

D. 9

Ans: B

EXPLANATION O1 has 2 items, O2 has 1 and O3 has 3: 2 + 1 + 3 = 6 (NULLs are not items).

Q65 · *Table-based | Normalisation*

MODERATE

Which of the following relations is in 1NF?

RELATION A

EmpID	Skills
E1	Java, SQL
E2	C

RELATION B

EmpID	Skill
E1	Java
E1	SQL
E2	C

RELATION C

EmpID	Skill1	Skill2
E1	Java	SQL
E2	C	NULL

A. Relation A

B. Relation B

C. Relation C

D. None of them

Ans: B

EXPLANATION Relation B has one skill per row; A has a list in one cell and C repeats Skill columns (a repeating group).

Q66 · *Error-identification | Normalisation*

EASY

In this relation, the cell that violates 1NF belongs to:

STUDENT

RollNo	Name	Languages
R1	Anu	Hindi
R2	Bala	English
R3	Chitra	Hindi, English
R4	Dev	Odia

A. Student R1

B. Student R2

C. Student R3

D. Student R4

Ans: C

EXPLANATION R3 has two languages in one cell, so the value is not atomic.

SECTION 05

Second Normal Form (2NF)

32 Questions · 9 Easy · 16 Moderate · 7 Difficult

Q67 · *Definition | Normalisation* EASY

A relation is in 2NF if it is:

A. In 1NF with no transitive dependency among non-key attributes

B. In 3NF with no repeating groups in any column

C. Only in 1NF, with every value atomic in each cell

D. In 1NF with no partial dependency on the key

Ans: D

EXPLANATION 2NF requires 1NF plus the absence of partial dependency.

Q68 · *Definition | Normalisation* EASY

Partial dependency occurs when:

A. A non-key attribute depends on part of a composite key

B. A key attribute depends on a non-key attribute

C. A non-key attribute depends on the whole primary key

D. A non-key attribute depends on another non-key attribute

Ans: A

EXPLANATION Partial dependency specifically involves dependency on only a portion of a composite key.

Q69 · *Concept | Normalisation* MODERATE

2NF is relevant mainly when a relation has:

A. A primary key that has a single attribute only

B. A table with only foreign keys and no primary key

C. A table in which no primary key has been defined

D. A composite (multi-attribute) primary key

Ans: D

EXPLANATION Partial dependency, and thus 2NF violations, can only occur with a composite key.

Q70 · *Table-based | Normalisation* MODERATE

Table key (StudentID, CourseID) -> StudentName, CourseName. StudentName depends only on StudentID and CourseName depends only on CourseID. This relation:

A. Satisfies 2NF, as each attribute depends on a key part

B. Violates 1NF, as the key has two attributes

C. Satisfies 3NF but not 2NF, due to a transitive dependency

D. Violates 2NF, due to partial dependency

Ans: D

EXPLANATION Both StudentName and CourseName depend on only part of the composite key, which is partial dependency.

Q71 · *Table-based | Normalisation* MODERATE

To convert the StudentID/CourseID table above into 2NF, you should:

A. Remove the CourseID attribute from the table entirely

B. Split it into Student, Course and Enrollment tables

C. Merge all attributes into one bigger table to avoid joins

D. Add more repeating columns for each course taken

Ans: B

EXPLANATION Removing partial dependency requires separating attributes that depend on only part of the key into their own relations.

Q72 · *Definition | Normalisation* MODERATE

2NF requires that every non-key attribute be:

A. Independent of the primary key of the relation entirely

B. Dependent on only a part of the composite primary key

C. Fully dependent on the whole primary key

D. Dependent on another non-key attribute of the relation

Ans: C

EXPLANATION Full functional dependency on the whole key is exactly what 2NF requires.

Q73 · *Concept | Normalisation* EASY

If a relation has only a single-attribute primary key, it automatically:

A. Satisfies 2NF if already in 1NF

B. Violates 1NF because the key is only one attribute

C. Cannot be normalized any further than 1NF

D. Violates 2NF because the key is only one attribute

Ans: A

EXPLANATION Partial dependency needs a composite key; with a single-attribute key, there is no possibility of partial dependency.

Q74 · *Concept | Normalisation* MODERATE

Which of the following is the correct relationship between 1NF and 2NF?

A. A relation must already be in 1NF to be in 2NF

B. A relation must already be in 3NF before it can be in 2NF

C. 2NF is a weaker requirement than 1NF and does not include it

D. 2NF has no relationship to 1NF; they are independent

Ans: A

EXPLANATION 2NF builds on top of 1NF; a relation is checked for 2NF only after confirming it satisfies 1NF.

Q75 · *Table-based | Normalisation* DIFFICULT

Key: (OrderID, ProductID); Quantity depends on both OrderID and ProductID; ProductName depends only on ProductID. This relation:

A. It violates 1NF, since Quantity is not an atomic value

B. It satisfies 3NF but violates 1NF because of repeated rows

C. It satisfies 2NF completely, since Quantity depends on the full key

D. It violates 2NF, as ProductName is partially dependent on the key

Ans: D

EXPLANATION ProductName depends on only part of the composite key (ProductID), which violates 2NF.

Q76 · *Concept | Normalisation* EASY

Removing partial dependency is the main goal of achieving:

A. 1NF

B. 2NF

C. 3NF

D. None of these

Ans: B

EXPLANATION The elimination of partial dependency is specifically the requirement that defines 2NF.

Q77 · *Definition | Normalisation* MODERATE

'Full functional dependency' (as needed to explain 2NF) means:

A. An attribute depends on a key of another table

B. An attribute depends on another non-key attribute only

C. An attribute depends on a part of the composite key

D. An attribute depends on the whole composite key

Ans: D

EXPLANATION Full functional dependency contrasts with partial dependency — it means dependence on the entire key.

Q78 · *Table-based | Normalisation* MODERATE

Key (StudentID, SubjectID) -> Marks, StudentName. If StudentName depends only on StudentID, this relation:

A. Is in 2NF because Marks depends on the full key

B. Is not in 2NF

C. Is in 3NF because no transitive dependency exists

D. Is not in 1NF because values are non-atomic

Ans: B

EXPLANATION StudentName's dependency on only StudentID (part of the composite key) is a 2NF violation.

Q79 · *Concept | Normalisation* EASY

A 2NF violation can be identified by looking for:

A. Non-key attributes depending on part of a composite key

B. Foreign keys that are missing from the table

C. Non-key attributes that depend on other non-key attributes

D. Non-atomic values stored in a column of the table

Ans: A

EXPLANATION This is precisely the definition of a partial dependency, which 2NF forbids.

Q80 · *Table-based | Normalisation* MODERATE

A relation with key (StudentID, CourseID) has only one non-key attribute, Grade, which depends on both parts of the key together. This relation:

A. Satisfies 2NF, as Grade needs the whole composite key

B. Violates 2NF, as Grade shows a partial dependency on one key part

C. Violates 1NF, as Grade has non-atomic values

D. Cannot be judged without knowing more attributes

Ans: A

EXPLANATION Since Grade depends on both StudentID and CourseID together, there is no partial dependency, so 2NF is satisfied.

Q81 · *Concept | Normalisation* MODERATE

Which scenario best demonstrates a partial dependency?

A. With key OrderID, OrderDate depends on the whole key OrderID

B. With key ProductID, ProductPrice depends on the whole key ProductID

C. With key EmployeeID, EmployeeName depends on the whole key EmployeeID

D. With key (OrderID, ProductID), CustomerName depends only on OrderID

Ans: D

EXPLANATION CustomerName depending on only part (OrderID) of the composite key is a partial dependency.

Q82 · *Concept | Normalisation* EASY

Simple relations with a single-column primary key and no repeating groups are usually already in:

A. 2NF, if they are in 1NF

B. Never in 1NF

C. Only in 3NF, never in 2NF

D. Never in 2NF

Ans: A

EXPLANATION With a single-attribute key, partial dependency cannot occur, so satisfying 1NF automatically satisfies 2NF.

Q83 · *Table-based | Normalisation* DIFFICULT

Table (StudentID, CourseID) -> Grade, InstructorName; InstructorName depends only on CourseID. What decomposition removes the 2NF violation?

A. Merge InstructorName into StudentID so every student has an instructor

B. Delete InstructorName from the database altogether to avoid repetition

C. Move (StudentID, Grade) into a separate relation and keep the rest unchanged

D. Move (CourseID, InstructorName) into a separate relation

Ans: D

EXPLANATION Moving InstructorName (which depends only on CourseID) into its own Course-based relation removes the partial dependency.

Q84 · *Concept | Normalisation* MODERATE

Which of these correctly explains why 2NF depends on the type of primary key?

A. 2NF applies only to relations that have no primary key

B. Partial dependency occurs equally with every type of key

C. 2NF has no connection with the primary key chosen

D. Partial dependency can exist only with a composite key

Ans: D

EXPLANATION A single-attribute key cannot be 'partially' depended upon, so 2NF concerns are specific to composite keys.

Q85 · *Table-based | Normalisation* MODERATE

Key (BranchID, EmployeeID) -> BranchLocation, EmployeeSalary. BranchLocation depends only on BranchID. This relation:

A. Satisfies 3NF

B. Violates 2NF

C. Satisfies 2NF

D. Violates 1NF

Ans: B

EXPLANATION BranchLocation's dependency on only BranchID (part of the key) is a partial dependency.

Q86 · *Concept | Normalisation* EASY

2NF is best described as eliminating:

A. Redundancy caused by attributes depending on other non-key attributes

B. Redundancy caused by non-atomic attribute values

C. Redundancy caused by attributes depending on part of a composite key

D. Redundancy caused by having too many tables

Ans: C

EXPLANATION 2NF specifically removes redundancy from partial dependency on a composite key.

Q87 · *Decomposition | Normalisation* MODERATE

Given key (StudentID, CourseID) with StudentName, CourseName, and Marks, where Marks depends on the full key while StudentName/CourseName depend on parts of it, the correct decomposition is:

A. Student(StudentID, CourseName, Marks), Course(CourseID, StudentName), Enrollment(StudentID, CourseID, Marks)

B. Student(StudentID, StudentName), Course(CourseID, CourseName), Enrollment(StudentID, CourseID, Marks)

C. One table holding StudentID, CourseID, StudentName, CourseName and Marks together without any split

D. Student(StudentID, StudentName, Marks) and Course(CourseID, CourseName) only

Ans: B

EXPLANATION Attributes are grouped by what they truly depend on: Student facts, Course facts, and the enrollment fact tied to both.

Q88 · *Concept | Normalisation* MODERATE

If a relation is in 2NF, it is guaranteed to also be in:

A. 1NF

B. Only 2NF itself, never 1NF

C. 3NF

D. None of these

Ans: A

EXPLANATION 2NF is defined as 1NF plus the absence of partial dependency, so 2NF always implies 1NF.

Q89 · *Concept | Normalisation* EASY

Which best defines the purpose of 2NF?

A. Every table must be free from foreign keys and composite keys

B. All attributes in every table must have atomic values only

C. Non-key attributes must depend on the whole key, not part of it

D. Non-key attributes must depend on other non-key attributes of the table

Ans: C

EXPLANATION This is the exact defining condition of Second Normal Form.

Q90 · *Table-based | Normalisation* DIFFICULT

Key (ProjectID, EmployeeID) -> HoursWorked, EmployeeDept. EmployeeDept depends only on EmployeeID. Identify the violation and correct table split.

A. 1NF violation; EmployeeDept is non-atomic, so split it into many columns

B. No violation exists here; the relation is already in 2NF, so no split is needed at all

C. Transitive dependency; Project(ProjectID, EmployeeDept) and Employee(EmployeeID, HoursWorked) tables are made

D. Partial dependency; Employee(EmployeeID, EmployeeDept) and Assignment(ProjectID, EmployeeID, HoursWorked)

Ans: D

EXPLANATION EmployeeDept depending only on part of the key (EmployeeID) is partial dependency, corrected by isolating it in an Employee relation.

Q91 · *Table-based | Normalisation* MODERATE

Key is (StudentID, CourseID). Which attribute causes the 2NF violation?

ENROLLMENT

StudentID	CourseID	StudentName	Grade
S1	C1	Asha	A
S1	C2	Asha	B
S2	C1	Ravi	A

A. None; the relation is already in 2NF as it stands

B. CourseID, because it is already part of the composite key

C. Grade, because it depends on both key attributes together

D. StudentName, because it depends only on StudentID

Ans: D

EXPLANATION StudentName repeats for each course of S1, showing it depends on only part of the key.

Q92 · *Table-based | Normalisation* MODERATE

Key is (OrderID, ProductID) and Qty depends on the whole key. The highest normal form satisfied is:

ORDERLINE

OrderID	ProductID	Qty
O1	P1	2
O1	P2	1
O2	P1	5

A. 3NF

B. 1NF only

C. 2NF only

D. Not even 1NF

Ans: A

EXPLANATION Atomic values, no partial dependency (Qty needs both) and no transitive dependency give 3NF.

Q93 · *Decomposition | Normalisation* DIFFICULT

Key is (OrderID, ProductID). Which decomposition gives 2NF?

ORDERDETAIL

OrderID	ProductID	ProductName	Qty
O1	P1	Pen	2
O1	P2	Book	1
O2	P1	Pen	5

A. Keep the single relation unchanged with all attributes together

B. Product(ProductID, Qty) and OrderDetail(OrderID, ProductID, ProductName)

C. OrderDetail(OrderID, ProductName) and Product(ProductID, OrderID, Qty)

D. Product(ProductID, ProductName) and OrderDetail(OrderID, ProductID, Qty)

Ans: D

EXPLANATION ProductName depends on ProductID alone, so it moves to a Product table; Qty stays with the full key.

Q94 · *Dependency-based | Normalisation* DIFFICULT

Key is (StudentID, SubjectID). Which attributes show a partial dependency?

RESULT

StudentID	SubjectID	StudentName	SubjectName	Marks
S1	M1	Asha	Maths	80
S1	M2	Asha	Physics	75
S2	M1	Ravi	Maths	66

A. StudentName and SubjectName

B. Marks together with StudentName

C. SubjectName alone, without StudentName

D. Marks alone, which depends on the whole key

Ans: A

EXPLANATION StudentName depends on StudentID and SubjectName on SubjectID; Marks needs both.

Q95 · *Dependency-based | Normalisation* DIFFICULT

From the rows, which dependency is evident (key is (StudentID, CourseID))?

ENROLLMENT

StudentID	CourseID	CourseName	Grade
S1	C1	DBMS	A
S2	C1	DBMS	B
S3	C2	Java	A
S1	C2	Java	C

A. CourseName depends on CourseID alone, a part of the key

B. Grade depends on CourseName rather than on the key

C. StudentID depends on CourseName in every row shown

D. CourseName depends on the whole key (StudentID, CourseID)

Ans: A

EXPLANATION Every row with C1 has DBMS, and every row with C2 has Java, whatever the student.

Q96 · *Concept | Normalisation* MODERATE

Does this relation (key is EmpID) satisfy 2NF?

EMPLOYEE

EmpID	EmpName	DeptID	DeptName
E1	Asha	D1	Sales
E2	Ravi	D1	Sales

A. It cannot be decided unless many more rows are available

B. No, DeptName depends on DeptID, which is only part of the key

C. Yes, a single-attribute key rules out any partial dependency

D. No, EmpName depends on only a part of the primary key

Ans: C

EXPLANATION 2NF is only about partial dependency, which needs a composite key; DeptName's issue belongs to 3NF.

Q97 · *Decomposition | Normalisation* EASY

Result(StudentID, CourseID, StudentName, CourseName, Marks) is decomposed into 2NF as Student(StudentID, StudentName), Course(CourseID, CourseName) and Result(StudentID, CourseID, Marks). How many relations result?

A. 2

B. 3

C. 4

D. 5

Ans: B

EXPLANATION Student, Course and the Result relation: three relations.

Q98 · *Statement-based | Normalisation* DIFFICULT

(i) 2NF needs the relation to be in 1NF. (ii) Partial dependency can occur with a single-attribute key. (iii) 2NF removes partial dependency. Which are correct?

A. (ii) and (iii) only

B. (i), (ii) and (iii)

C. (i) and (ii) only

D. (i) and (iii) only

Ans: D

EXPLANATION Partial dependency needs a composite key, so (ii) is false.

SECTION 06

Third Normal Form (3NF)

28 Questions · 8 Easy · 14 Moderate · 6 Difficult

Q99 · *Definition | Normalisation* EASY

A relation is in 3NF if it is:

A. In 2NF with no transitive dependency

B. In 2NF with no partial dependency on the key

C. In 2NF with only atomic values in cells

D. In 1NF only, with every value atomic

Ans: A

EXPLANATION 3NF requires 2NF plus the elimination of transitive dependency.

Q100 · *Definition | Normalisation* EASY

Transitive dependency occurs when:

A. A non-key attribute depends on part of a composite primary key

B. A non-key attribute depends directly on the whole primary key

C. A key attribute depends on a non-key attribute of the table

D. A non-key attribute depends on another non-key attribute

Ans: D

EXPLANATION This describes transitive dependency: key -> A -> B, where B depends on A rather than the key directly.

Q101 · *Table-based | Normalisation* MODERATE

Key is EmpID. Why does this relation violate 3NF?

EMPLOYEE

EmpID	DeptID	DeptName
E1	D10	Sales
E2	D10	Sales
E3	D20	HR

A. DeptName depends on only a part of the composite primary key

B. DeptName depends on non-key DeptID, a transitive dependency

C. The relation has no violation since EmpID is the primary key

D. DeptID holds more than one value in a single cell of the row

Ans: B

EXPLANATION EmpID -> DeptID -> DeptName is a transitive chain through a non-key attribute.

Q102 · *Table-based | Normalisation* MODERATE

To remove the transitive dependency in the EmployeeID/DeptID/DeptName example, you should:

A. Delete the DeptID column from the table completely

B. Make a Department table with DeptID and DeptName

C. Merge DeptName into the EmployeeID column

D. Keep all three attributes together in one table

Ans: B

EXPLANATION Separating Department information into its own relation removes the transitive dependency.

Q103 · *Concept | Normalisation* EASY

The main goal of achieving 3NF is to eliminate:

A. Transitive dependency between non-key attributes

B. Partial dependency of non-key attributes on a composite key

C. All foreign key relationships between the tables

D. Non-atomic attribute values in a column

Ans: A

EXPLANATION 3NF specifically targets transitive dependency, distinguishing it from 2NF's partial dependency focus.

Q104 · *Concept | Normalisation* MODERATE

Which of the following correctly relates 2NF and 3NF?

A. 3NF is a weaker form than 2NF, so 3NF is always satisfied first

B. A relation must be in 2NF before it is checked for 3NF

C. 2NF and 3NF are independent forms, checked in any order

D. A relation must be in 3NF before it is checked for 2NF

Ans: B

EXPLANATION 3NF builds upon 2NF, just as 2NF builds upon 1NF.

Q105 · *Table-based | Normalisation* MODERATE

Key StudentID -> HostelID -> HostelWarden (HostelWarden depends on HostelID). This is an example of:

A. Transitive dependency

B. Repeating group of columns

C. No dependency among attributes

D. Partial dependency on the key

Ans: A

EXPLANATION HostelWarden depends on HostelID, a non-key attribute, rather than directly on StudentID — a transitive dependency.

Q106 · *Concept | Normalisation*

EASY

A relation with a single-attribute key that is already in 2NF may still violate 3NF if:

A. The primary key is composite instead of single

B. Some attribute values are not atomic in the relation

C. A non-key attribute depends on another non-key attribute

D. A non-key attribute depends on only part of the primary key

Ans: C

EXPLANATION Transitive dependency (violating 3NF) can occur even with a single-attribute key, unlike partial dependency.

Q107 · *Table-based | Normalisation*

DIFFICULT

Key is EmpID, and each project has one manager. Which statement is correct?

EMPLOYEE

EmpID	EmpName	ProjectID	ProjectManager
E1	Ravi	P1	Sharma
E2	Neha	P1	Sharma
E3	Amit	P2	Verma

A. It is in 2NF but not 3NF; ProjectManager depends on ProjectID

B. It is in 3NF, since EmpID is a single-attribute primary key

C. It is not in 2NF, since ProjectManager is partially dependent on EmpID

D. It is not in 1NF, since ProjectManager is repeated in many rows

Ans: A

EXPLANATION A single key rules out partial dependency (2NF holds), but EmpID -> ProjectID -> ProjectManager is transitive.

Q108 · *Concept | Normalisation*

MODERATE

Removing a transitive dependency typically involves:

A. Making the dependent attribute the new primary key

B. Deleting the dependent attribute from the database for good

C. Merging all the attributes into a single relation

D. Moving the dependent attributes to a separate relation

Ans: D

EXPLANATION The standard fix is decomposition — moving the dependent attributes into their own relation.

Q109 · *Concept | Normalisation*

EASY

Which best defines 3NF?

A. Every attribute must hold several values to avoid creating more tables

B. Every attribute must be part of the composite primary key

C. Every non-key attribute depends on part of the primary key only

D. Every non-key attribute depends directly on the primary key alone

Ans: D

EXPLANATION This directness-of-dependency requirement is the essence of 3NF.

Q110 · *Table-based | Normalisation* MODERATE

Key is OrderID. Which statement about this relation is correct?

ORDERS

OrderID	CustomerID	CustomerCity
O1	C1	Pune
O2	C1	Pune
O3	C2	Delhi

A. It is in 3NF but not in 2NF

B. It is in 2NF but not in 3NF

C. It is not in 1NF because of repeated values

D. It is in 1NF but not in 2NF because of a partial dependency

Ans: B

EXPLANATION No partial dependency is possible with a single key, but CustomerCity depends on CustomerID, a non-key attribute.

Q111 · *Concept | Normalisation* MODERATE

If a relation is in 3NF, it is guaranteed to also be in:

A. Only 2NF, not 1NF

B. 2NF and 1NF

C. Only 1NF, not 2NF

D. Neither 1NF nor 2NF

Ans: B

EXPLANATION 3NF is defined as 2NF (which requires 1NF) plus no transitive dependency, so 3NF implies both.

Q112 · *Decomposition | Normalisation* DIFFICULT

Key StudentID -> CityID -> CityPincode. Which correct decomposition removes the transitive dependency?

A. Student(StudentID, CityID) and City(CityID, CityPincode)

B. Student(StudentID, CityPincode) and City(CityID, StudentID)

C. One combined table holding all three attributes together

D. Student(CityID, CityPincode) and City(StudentID, CityID)

Ans: A

EXPLANATION Separating City-related facts into their own relation keyed by CityID removes the transitive chain.

Q113 · *Concept | Normalisation* EASY

A transitive dependency involves at least:

A. Only key attributes of a composite primary key

B. Three attributes: the key and two non-key attributes

C. Two attributes: the key and one non-key attribute depending on it

D. A single attribute that is repeated in many rows

Ans: B

EXPLANATION A transitive dependency needs a chain: Key -> A -> B, involving three attributes.

Q114 · *Decomposition | Normalisation* MODERATE

Key is BookID. Which decomposition brings the relation into 3NF without losing the link between books and publishers?

BOOK

BookID	PublisherID	PublisherCountry
B1	P1	India
B2	P1	India
B3	P2	UK

A. Keep one single table having all three attributes together

B. Book(BookID, PublisherID) and Publisher(PublisherID, PublisherCountry)

C. Book(BookID) and Publisher(PublisherID, PublisherCountry) with no link between them

D. Book(BookID, PublisherCountry) and Publisher(PublisherID, BookID)

Ans: B

EXPLANATION PublisherCountry moves to a table keyed by PublisherID, while Book keeps PublisherID as the link.

Q115 · *Comparison | Normalisation* MODERATE

Which statement about 2NF and 3NF is TRUE?

A. Both 2NF and 3NF are concerned only with atomic values

B. Neither 2NF nor 3NF is concerned with any dependency

C. 2NF removes transitive dependency; 3NF removes partial dependency

D. 2NF removes partial dependency; 3NF removes transitive dependency

Ans: D

EXPLANATION This correctly distinguishes the focus of each normal form.

Q116 · *Table-based | Normalisation* MODERATE

Key is EmpID. What is the correct fix to reach 3NF?

EMPLOYEE

EmpID	EmpName	Designation	Allowance
E1	Ram	Manager	5000
E2	Sita	Clerk	2000
E3	Hari	Manager	5000

A. Create Employee(EmpID, Allowance) and delete Designation from the Employee table

B. Make Designation the primary key of Employee instead of EmpID

C. Create Designation(Designation, Allowance); keep Designation in Employee

D. Merge EmpName and Allowance into one column to remove the dependency

Ans: C

EXPLANATION Allowance depends on Designation, so it moves to its own table keyed by Designation.

Q117 · *Concept | Normalisation* EASY

3NF ensures that non-key attributes are dependent:

A. On no attribute of the relation at all

B. Only on the key, not on other non-key attributes

C. On a part of the composite key only, and not the whole key

D. Only on other non-key attributes of the same relation

Ans: B

EXPLANATION This direct-dependency-on-the-key-only rule is what 3NF enforces.

Q118 · *Table-based | Normalisation* DIFFICULT

Key: OrderID -> ProductID -> ProductCategory -> CategoryDiscount. Which pair of attributes exhibits transitive dependency relative to OrderID?

A. ProductCategory and CategoryDiscount, via ProductID

B. ProductID, which depends directly on OrderID

C. OrderID and ProductID, which form the direct key relationship

D. None of them shows a chain of dependency at all

Ans: A

EXPLANATION ProductCategory and CategoryDiscount both depend on OrderID only through the chain via ProductID.

Q119 · *Concept | Normalisation* MODERATE

Achieving 3NF after 2NF mainly involves examining:

A. Whether the table has a primary key at all

B. Whether attribute values are atomic

C. Dependencies among the non-key attributes themselves

D. Dependencies of non-key attributes on only part of the key

Ans: C

EXPLANATION 3NF analysis focuses on whether non-key attributes depend on each other rather than the key directly.

Q120 · *Table-based | Normalisation* MODERATE

Key is StudentID. Assuming it is in 1NF, the highest normal form satisfied is:

STUDENT

StudentID	AdvisorID	AdvisorOffice
S1	A1	Room 5
S2	A1	Room 5
S3	A2	Room 9

A. 2NF

B. 1NF only

C. 3NF

D. None of these

Ans: A

EXPLANATION No partial dependency (single key), but AdvisorOffice depends on AdvisorID, so 3NF fails.

Q121 · *Concept | Normalisation* EASY

Which of these is an example of removing a transitive dependency?

A. Splitting EmployeeName into first and last name columns

B. Combining two composite keys into one composite key

C. Moving DeptName out of Employee into a separate Department table

D. Merging DeptName further into the Employee table with all other columns

Ans: C

EXPLANATION This is the classic transitive-dependency fix — isolating the dependent non-key attribute in its own relation.

Q122 · *Decomposition | Normalisation* DIFFICULT

Given key EmployeeID -> ManagerID -> ManagerName, which of these is correctly in 3NF?

A. Employee(EmployeeID, ManagerID) and Manager(ManagerID, ManagerName)

B. A single relation with ManagerName chosen as the primary key

C. Employee(EmployeeID, ManagerName) and Manager(ManagerID) with no names

D. Employee(EmployeeID, ManagerID, ManagerName) kept as a single relation

Ans: A

EXPLANATION Splitting so ManagerName depends directly on its own key (ManagerID) removes the transitive dependency.

Q123 · *Decomposition | Normalisation* MODERATE

Key is EmpID. How many relations result when this relation is decomposed into 3NF by removing the transitive dependency?

EMPLOYEE

EmpID	DeptID	DeptName
E1	D1	Sales
E2	D1	Sales
E3	D2	HR

A. 1

B. 2

C. 3

D. 4

Ans: B

EXPLANATION Employee(EmpID, DeptID) and Department(DeptID, DeptName).

Q124 · *Dependency-based | Normalisation* MODERATE

Key is RollNo and each city has one pincode. Which attribute is transitively dependent on RollNo?

STUDENT

RollNo	Name	City	Pincode
1	Asha	Pune	411001
2	Ravi	Pune	411001
3	Neel	Delhi	110001

A. Name

B. City

C. Pincode, through City

D. None of these

Ans: C

EXPLANATION RollNo -> City -> Pincode is the transitive chain.

Q125 · *Decomposition | Normalisation* DIFFICULT

Key is OrderID. Which decomposition gives 3NF?

ORDER

OrderID	CustomerID	CustomerName	CustomerCity	OrderDate
O1	C1	Asha	Pune	01-Jun
O2	C1	Asha	Pune	03-Jun
O3	C2	Ravi	Delhi	04-Jun

A. Order(OrderID, OrderDate) and Customer(CustomerID, CustomerName)

B. Order(OrderID, CustomerName, OrderDate, CustomerCity) and Customer(CustomerID, CustomerName)

C. Order(OrderID, CustomerID, OrderDate); Customer(CustomerID, CustomerName, CustomerCity)

D. Order(OrderID, CustomerID, CustomerName) and Customer(CustomerCity, OrderDate)

Ans: C

EXPLANATION CustomerName and CustomerCity depend on CustomerID, so they move out; OrderDate depends on OrderID.

Q126 · *Table-based | Normalisation* DIFFICULT

Which of the following relations is in 2NF but not in 3NF?

R1 (KEY SID)

SID	Name	Age
1	Asha	17
2	Ravi	18

R2 (KEY EID)

EID	DID	DName
E1	D1	Sales
E2	D1	Sales

R3 (KEY SID, CID)

SID	CID	Mark	CName
1	C1	80	DBMS
2	C1	75	DBMS

A. Relation R1

B. Relation R2

C. Relation R3

D. None of them

Ans: B

EXPLANATION R1 is in 3NF; R3 (key SID, CID) has CName depending on CID, a partial dependency (1NF only); R2 has a transitive dependency EID -> DID -> DName.

SECTION 07

Rules and Examples of 1NF, 2NF and 3NF

25 Questions · 3 Easy · 18 Moderate · 4 Difficult

Q127 · *Table-based | Normalisation* MODERATE

The highest normal form satisfied by this relation is:

STUDENT

StudentID	StudentName	Subjects
S1	Anil	Math, Science
S2	Bina	English

A. 1NF only, since it has a primary key and few columns

B. Not even 1NF, since some cells hold multiple values

C. 2NF, since no non-key attribute depends on part of a key

D. 3NF, since there is no dependency among non-key attributes

Ans: B

EXPLANATION The Subjects cell for S1 holds two values, so atomicity (1NF) fails.

Q128 · *Table-based | Normalisation* MODERATE

Table key (StudentID, CourseID) -> StudentName, Marks, where StudentName depends only on StudentID. Highest normal form satisfied:

A. 1NF only

B. Not even in 1NF

C. 3NF

D. 2NF

Ans: A

EXPLANATION All values are atomic (1NF holds), but StudentName's partial dependency on StudentID means 2NF fails.

Q129 · *Table-based | Normalisation* MODERATE

Key is EmpID. The highest normal form satisfied is:

EMPLOYEE

EmpID	EmpName	DeptID	DeptName
E1	Asha	D1	Sales
E2	Ravi	D1	Sales
E3	Neel	D2	HR

A. 3NF

B. 2NF

C. Not even 1NF

D. 1NF only

Ans: B

EXPLANATION Values are atomic and the key is single (2NF holds), but DeptName depends on DeptID, a non-key attribute.

Q130 · *Table-based | Normalisation* MODERATE

Key is BookID and no attribute depends on another non-key attribute. The highest normal form satisfied is:

BOOK

BookID	Title	Author	Price
B1	DBMS Basics	Korth	450
B2	Networks	Forouzan	520

A. 3NF

B. 1NF only

C. Not even 1NF

D. 2NF only

Ans: A

EXPLANATION Atomic values, no partial dependency and no transitive dependency give 3NF (which includes 2NF and 1NF).

Q131 · *Rule-based | Normalisation* EASY

Which rule is specific to 1NF and not required directly by 2NF or 3NF?

A. Presence of a composite primary key in the table

B. Absence of transitive dependency among attributes

C. Absence of partial dependency on the key

D. Atomicity of every attribute value

Ans: D

EXPLANATION Atomicity is the defining and distinguishing rule of 1NF.

Q132 · *Rule-based | Normalisation* MODERATE

Identify the rule violated: Table key (OrderID, ItemID) -> ItemName, where ItemName depends only on ItemID.

A. 1NF rule — atomicity

B. No rule is violated

C. 3NF rule — transitive dependency

D. 2NF rule — partial dependency

Ans: D

EXPLANATION ItemName depending on only part of the composite key is a textbook partial dependency, violating 2NF.

Q133 · *Rule-based | Normalisation* MODERATE

Identify the rule violated: Table StudentID -> HostelID -> HostelFee, where HostelFee depends on HostelID.

A. 3NF rule — transitive dependency

B. 2NF rule — partial dependency on a composite key

C. 1NF rule — atomicity of attribute values

D. No rule is violated by this table

Ans: A

EXPLANATION HostelFee depends on HostelID (non-key), not directly on StudentID, which is a transitive dependency.

Q134 · *Comparison | Normalisation* MODERATE

Compared to 1NF, 2NF additionally requires:

A. Only atomic values in every attribute of the relation

B. A single-attribute primary key in every relation

C. No transitive dependency among the non-key attributes

D. No partial dependency on a composite key

Ans: D

EXPLANATION 2NF adds the partial-dependency rule on top of the atomicity requirement of 1NF.

Q135 · *Comparison | Normalisation* MODERATE

Compared to 2NF, 3NF additionally requires:

A. Removal of all foreign keys from every relation

B. No partial dependency on a composite key attribute

C. No transitive dependency among non-key attributes

D. Only atomic values in every attribute of the table

Ans: C

EXPLANATION 3NF adds the transitive-dependency rule on top of 2NF's partial-dependency rule.

Q136 · *Comparison | Normalisation* MODERATE

A relation in 3NF but hypothetically not in 1NF is:

A. Only possible when the relation has a single-attribute key

B. Impossible, since 3NF always implies 1NF

C. Common in most databases that use composite keys

D. Only possible when the relation has composite keys

Ans: B

EXPLANATION Because 3NF requires 2NF, which requires 1NF, a relation cannot be in 3NF without also being in 1NF.

Q137 · *Table-based | Normalisation* DIFFICULT

Table key (ProjectID, EmployeeID) -> HoursWorked, EmployeeName, ProjectManager. EmployeeName depends only on EmployeeID; ProjectManager depends only on ProjectID. Which normal form does this satisfy?

A. None, not even 1NF

B. 3NF

C. 1NF only

D. 2NF

Ans: C

EXPLANATION Values are atomic (1NF), but both EmployeeName and ProjectManager show partial dependency, so 2NF fails.

Q138 · *Rule-based | Normalisation* EASY

Determining the highest normal form of a relation requires checking, in order:

A. Only atomicity needs to be checked and the rest is irrelevant

B. Transitive dependency, then atomicity, then partial dependency

C. Partial dependency, then transitive dependency, then atomicity

D. Atomicity, then partial dependency, then transitive dependency

Ans: D

EXPLANATION Normal forms are checked in increasing order of strictness: 1NF, then 2NF, then 3NF.

Q139 · *Table-based | Normalisation* MODERATE

Table: EmployeeID -> EmployeeName, Salary, with a single-attribute key and no attribute depending on another non-key attribute. This relation is in:

A. 1NF only

B. 2NF only

C. 3NF

D. None of these normal forms

Ans: C

EXPLANATION Atomic values, no possible partial dependency, and no transitive dependency means the relation satisfies 3NF.

Q140 · *Comparison | Normalisation* MODERATE

Which of these correctly compares 1NF and 3NF?

A. 1NF needs only atomic values; 3NF also removes partial and transitive dependency

B. 1NF is a stricter normal form than 3NF in every respect

C. 1NF removes transitive dependency; 3NF removes non-atomic values only

D. 1NF and 3NF both enforce exactly the same single rule on tables

Ans: A

EXPLANATION 3NF is a cumulative, stricter condition that includes 1NF's rule plus the additional dependency rules.

Q141 · *Decomposition | Normalisation* DIFFICULT

Table key (StudentID, CourseID) -> Grade, StudentName, CourseName; CourseName depends on CourseID and StudentName on StudentID. The correct 3NF decomposition is:

A. Student(StudentID, CourseName, Grade), Course(CourseID, StudentName), Enrollment(StudentID, CourseID, Grade)

B. Enrollment(StudentID, CourseID, StudentName, CourseName, Grade) as a single table

C. Student(StudentID, StudentName), Course(CourseID, CourseName), Enrollment(StudentID, CourseID, Grade)

D. One combined table holding all five attributes together without any split at all

Ans: C

EXPLANATION Each attribute is placed with the key it truly depends on, removing both partial dependency and redundancy.

Q142 · *Table-based | Normalisation* MODERATE

Table: EmployeeID | EmployeeName | ProjectID | ProjectName | ProjectBudget, EmployeeID as the single-attribute key; ProjectName and ProjectBudget depend on ProjectID (not directly on EmployeeID). Highest normal form:

A. Not even in 1NF

B. 2NF

C. 3NF

D. 1NF only

Ans: B

EXPLANATION No partial dependency is possible with a single-attribute key, but ProjectName/ProjectBudget transitively depend on EmployeeID via ProjectID, so 3NF fails.

Q143 · *Concept | Normalisation* EASY

A relation satisfying 3NF is generally considered:

A. Impossible to query because every fact is split into tables

B. Automatically the fastest design possible for every use case

C. Well structured, with little redundancy for most purposes

D. Poorly structured, with the maximum amount of redundancy

Ans: C

EXPLANATION 3NF is widely regarded as a good practical balance between redundancy elimination and design simplicity.

Q144 · *Comparison | Normalisation* MODERATE

Which statement about 1NF, 2NF, and 3NF is correct?

A. 3NF has fewer rules to satisfy than 1NF has

B. Each higher form replaces the rules of the lower forms

C. The three forms are independent of one another entirely

D. Each higher form includes the rules of the lower forms

Ans: D

EXPLANATION The normal forms are cumulative — 3NF includes the 1NF and 2NF rules plus its own additional rule.

Q145 · *Table-based | Normalisation* MODERATE

Key is (OrderID, ProductID). The highest normal form satisfied is:

ORDERLINE

OrderID	ProductID	ProductName	Qty
O1	P1	Pen	2
O2	P1	Pen	5
O2	P2	Book	1

A. Not even 1NF

B. 2NF

C. 1NF only

D. 3NF

Ans: C

EXPLANATION Values are atomic, but ProductName depends on ProductID alone, a partial dependency, so 2NF fails.

Q146 · *Rule-based | Normalisation* MODERATE

Which rule is violated by this relation?

FACULTY

FacID	FacName	Phones
F1	Rao	98111, 98222
F2	Iyer	97000

A. 1NF, as Phones holds more than one value

B. 2NF, because a non-key attribute depends on part of the key

C. No rule is violated, since every row has a primary key

D. 3NF, because a non-key attribute depends on another non-key attribute

Ans: A

EXPLANATION A cell with 'Phone1, Phone2' is not atomic.

Q147 · *Rule-based | Normalisation* MODERATE

Key is PatID. Which rule is violated?

PATIENT

PatID	Ward	WardNurse
P1	W1	Mary
P2	W1	Mary
P3	W2	John

A. 2NF, because WardNurse depends on only a part of the key

B. 1NF, because the Ward value is repeated in many rows

C. No rule is violated because PatID is a single-attribute key

D. 3NF, as WardNurse depends on non-key Ward

Ans: D

EXPLANATION PatID -> Ward -> WardNurse is a transitive dependency.

Q148 · *Rule-based | Normalisation* DIFFICULT

Key is (MemberID, BookID). Which rule is violated?

ISSUE

MemberID	BookID	IssueDate	MemberName
M1	B1	01-Jun	Asha
M1	B2	05-Jun	Asha
M2	B1	07-Jun	Ravi

A. 3NF; there is a transitive dependency among attributes

B. 2NF; MemberName depends on MemberID alone

C. No rule is violated, as the relation is fully normalized

D. 1NF; the value of MemberID repeats in several rows

Ans: B

EXPLANATION MemberName is determined by part of the key, which is a partial dependency.

Q149 · *Scenario-based | Normalisation* MODERATE

A relation is in 1NF but has a partial dependency. The appropriate normalization step is:

A. Decompose it to remove the transitive dependency (to reach 3NF)

B. Decompose it to remove the partial dependency (to reach 2NF)

C. Convert its values to atomic form (to reach 1NF)

D. No step is needed

Ans: B

EXPLANATION Removing partial dependency is exactly what 2NF requires.

Q150 · *Scenario-based | Normalisation* MODERATE

A relation is in 2NF but has a transitive dependency. The appropriate next step is:

A. Split its cells into atomic values to reach 1NF

B. Merge it with another relation to combine all their attributes into one

C. Move the transitively dependent attributes to a separate relation (3NF)

D. Remove the partial dependency on the composite key to reach 2NF again

Ans: C

EXPLANATION Transitive dependency is removed to reach 3NF.

Q151 · *Table-based | Normalisation* DIFFICULT

Key is EmpID. If every row with DeptID D1 shows the same DeptName and DeptHead, the highest normal form is:

EMPLOYEE

EmpID	DeptID	DeptName	DeptHead
E1	D1	Sales	Rao
E2	D1	Sales	Rao
E3	D2	HR	Iyer

A. Not even 1NF

B. 2NF

C. 3NF

D. 1NF only

Ans: B

EXPLANATION DeptName and DeptHead depend on DeptID, a non-key attribute, so 3NF fails; the single key keeps 2NF.

SECTION 08

Anomalies

27 Questions · 8 Easy · 16 Moderate · 3 Difficult

Q152 · *Definition | Normalisation* EASY

A data anomaly in a database refers to:

A. A copy of a table kept in a separate place for safety

B. A rule that forces every column to hold one value

C. A problem caused by poor table design or redundancy

D. A special type of primary key made of several attributes

Ans: C

EXPLANATION Anomalies are problems caused by redundant or poorly structured data.

Q153 · *Definition | Normalisation* EASY

An insertion anomaly occurs when:

A. The same fact is changed in many places by mistake

B. Some data cannot be added without other unrelated data

C. A row is deleted along with other useful data

D. Data can always be added with no restriction on any attribute

Ans: B

EXPLANATION An insertion anomaly happens when the table structure forces unrelated data to be present before a valid insertion.

Q154 · *Definition | Normalisation* EASY

An update anomaly occurs when:

A. A fact repeated in many rows is updated in only some of them

B. New data cannot be inserted because a key value is missing

C. Data is deleted along with other unrelated data in the same row

D. A row is accidentally duplicated while inserting a new record

Ans: A

EXPLANATION If a fact is repeated across rows, updating it everywhere correctly is error-prone — an update anomaly.

Q155 · *Definition | Normalisation* EASY

A deletion anomaly occurs when:

A. Inserting a record causes older records to be deleted automatically

B. Deleting one record also removes other useful information

C. Deleting a record leaves every other record unaffected and complete

D. Updating a record deletes unrelated rows from other tables

Ans: B

EXPLANATION A deletion anomaly is the unintended loss of unrelated but still-needed data when a row is deleted.

Q156 · *Table-based | Normalisation* MODERATE

Key is (StudentID, CourseID). If student S3 withdraws and her row is deleted, the details of Java and Iyer vanish. This is a:

ENROLLMENT

StudentID	CourseID	CourseName	Instructor
S1	C1	DBMS	Rao
S2	C1	DBMS	Rao
S3	C2	Java	Iyer

A. Update anomaly

B. Insertion anomaly

C. No anomaly

D. Deletion anomaly

Ans: D

EXPLANATION The course data existed only in the deleted row, so useful information is lost.

Q157 · *Table-based | Normalisation* MODERATE

In the same table, if a course's Instructor changes, every row for that course must be updated, or the data becomes inconsistent. This illustrates:

A. Deletion anomaly

B. Update anomaly

C. No anomaly at all

D. Insertion anomaly

Ans: B

EXPLANATION Needing to change the same fact in multiple places, with a risk of missing some rows, is an update anomaly.

Q158 · *Table-based | Normalisation* MODERATE

In the same table, a new course cannot be added until at least one student enrolls in it (because StudentID is part of the key). This illustrates:

A. Deletion anomaly, from losing course data on removal

B. No anomaly, as the key allows every insert

C. Insertion anomaly

D. Update anomaly, from changing repeated course data

Ans: C

EXPLANATION The design forces unrelated data (a student) to exist before valid course data can be inserted.

Q159 · *Concept | Normalisation* MODERATE

The root cause of most anomalies in a poorly designed table is:

A. Excessive normalization of the tables

B. Too many separate tables joined by keys

C. Overuse of primary keys in each table

D. Unnecessary data redundancy

Ans: D

EXPLANATION Redundant storage of the same fact across many rows is what leads to insertion, update, and deletion anomalies.

Q160 · *Concept | Normalisation* EASY

Normalization helps reduce anomalies mainly by:

A. Combining unrelated tables into one

B. Removing all primary keys

C. Increasing redundant storage of the same fact

D. Reducing redundant storage of the same fact

Ans: D

EXPLANATION Since anomalies stem from redundancy, reducing redundancy through normalization reduces anomalies.

Q161 · *Concept | Normalisation* MODERATE

Which of the following best distinguishes an insertion anomaly from an update anomaly?

A. Insertion anomaly blocks adding valid data; update anomaly risks inconsistency

B. Insertion anomaly means data is lost on deletion; update anomaly means data cannot be added

C. Insertion and update anomalies are exactly the same problem under two names

D. Insertion anomaly occurs only while deleting rows; update anomaly only while inserting

Ans: A

EXPLANATION These are two distinct problems: one about being unable to insert, the other about inconsistent updates.

Q162 · *Concept | Normalisation* MODERATE

Which of the following best distinguishes an update anomaly from a deletion anomaly?

A. Update anomaly occurs only when new rows are being inserted

B. Update: inconsistency from repeated data; deletion: unintended loss of data

C. Both anomalies are triggered only by insert operations on new rows

D. Deletion anomaly never causes any loss of useful information

Ans: B

EXPLANATION Update anomalies concern keeping duplicated facts consistent; deletion anomalies concern accidentally losing information.

Q163 · *Table-based | Normalisation* DIFFICULT

Employee E3 leaves the company and her row is deleted. Which anomaly occurs?

EMPLOYEE

EmpID	DeptID	DeptName	DeptLocation
E1	D1	Sales	Pune
E2	D1	Sales	Pune
E3	D2	HR	Delhi

A. Update anomaly, because DeptName must change in several rows

B. Deletion anomaly, as HR department details exist only in E3's row

C. Insertion anomaly, because no new department can be added to the table

D. No anomaly, because E3's data was meant to be deleted anyway

Ans: B

EXPLANATION Removing the only row for D2 also removes the facts about D2 (HR, Delhi).

Q164 · *Concept | Normalisation* EASY

Anomalies are more likely to occur in:

A. Unnormalized or poorly normalized tables

B. Well-normalized tables that are in 3NF

C. Tables that have a proper primary key and atomic values

D. Tables in which every fact is stored only once

Ans: A

EXPLANATION Redundancy, which causes anomalies, is far more common in unnormalized or under-normalized designs.

Q165 · *Concept | Normalisation* MODERATE

Which of these is a correct example of an insertion anomaly?

A. A new product cannot be added until it is ordered once

B. A product's category is kept once in a Category table

C. A product is removed and no other data is lost

D. A product's price is changed correctly in a single row only

Ans: A

EXPLANATION Requiring an unrelated event before valid data can be stored is a classic insertion anomaly.

Q166 · *Concept | Normalisation* MODERATE

Which of these is a correct example of a deletion anomaly?

A. Updating a customer's phone number in one place only, with no other copies

B. Adding a new customer who has not yet placed any order, which is not possible

C. Deleting a customer's only order also removes the customer's contact details

D. Deleting an order simply removes that order's record and nothing else

Ans: C

EXPLANATION Losing the customer's information as a side effect of deleting an unrelated order record is a deletion anomaly.

Q167 · *Concept | Normalisation* MODERATE

Which of these is a correct example of an update anomaly?

A. A customer's address stored in every order row must be changed in many rows

B. Deleting a customer also deletes all the orders of that customer

C. A customer's address is stored just once in a Customers table

D. A new order is added for an existing customer without any address change

Ans: A

EXPLANATION Needing to update the same repeated fact in multiple rows, risking some being missed, is the definition of an update anomaly.

Q168 · *Concept | Normalisation* EASY

Which of the following is NOT typically classified as one of the three classic anomalies?

A. Insertion anomaly

B. Update anomaly

C. Query anomaly

D. Deletion anomaly

Ans: C

EXPLANATION The three classic anomalies are insertion, update, and deletion anomalies; 'query anomaly' is not a standard term.

Q169 · *Concept | Normalisation* MODERATE

The relationship between redundancy and anomalies is best described as:

A. Redundancy always removes anomalies from the tables

B. Anomalies are the cause; redundancy is the result

C. Redundancy and anomalies have no link with each other

D. Redundancy is the cause; anomalies are the result

Ans: D

EXPLANATION Redundant data storage is the root cause that leads to insertion, update, and deletion anomalies.

Q170 · *Table-based | Normalisation* MODERATE

In this relation, author Rao is shown with two different countries. Which problem does this illustrate?

BOOK

BookID	Title	AuthorName	AuthorCountry
B1	DBMS	Rao	India
B2	Java	Rao	UK
B3	Python	Smith	USA

A. Update anomaly

B. Violation of 1NF

C. Insertion anomaly

D. Deletion anomaly

Ans: A

EXPLANATION AuthorCountry is stored once per book, so a change applied to only one row leaves inconsistent data.

Q171 · *Concept | Normalisation* EASY

Normalization reduces which of the following?

A. The likelihood of insertion, update, and deletion anomalies

B. The accuracy of the data stored in the tables

C. The number of valid primary keys that a table can contain in total

D. The number of relationships that can exist between tables

Ans: A

EXPLANATION By removing redundancy, normalization directly reduces the likelihood of all three classic anomalies.

Q172 · *Concept | Normalisation* MODERATE

In an unnormalized order table storing product details for every order line, adding a brand-new product with no orders yet is difficult. This scenario is an example of:

A. 1NF violation only

B. Deletion anomaly

C. Insertion anomaly

D. Update anomaly

Ans: C

EXPLANATION The table structure requires an order to exist before a product can be recorded.

Q173 · *Concept | Normalisation* MODERATE

Which statement correctly compares the three anomaly types?

A. All three anomalies describe exactly the same underlying problem

B. Update anomaly occurs only when the table has a composite primary key

C. Insertion blocks valid additions, update risks inconsistency, deletion loses data

D. Insertion anomaly causes data loss and deletion anomaly blocks new additions

Ans: C

EXPLANATION This correctly captures the distinct nature of each of the three classic anomalies.

Q174 · *Table-based | Normalisation* MODERATE

A new course C9 with no students yet cannot be added to this relation (key is (StudentID, CourseID)). This is a:

ENROLLMENT

StudentID	CourseID	CourseName
S1	C1	DBMS
S2	C1	DBMS
S1	C2	Java

A. Insertion anomaly

B. No anomaly, since the key is composite

C. Deletion anomaly, from losing student data

D. Update anomaly, from repeated course data

Ans: A

EXPLANATION Course data cannot be stored without a student, since StudentID is part of the key.

Q175 · *Table-based | Normalisation* MODERATE

Department D1 is renamed 'Marketing'. How many rows must be updated in this relation to stay consistent?

EMPLOYEE

EmpID	DeptID	DeptName
E1	D1	Sales
E2	D1	Sales
E3	D2	HR
E4	D1	Sales

A. 1

B. 2

C. 3

D. 4

Ans: C

EXPLANATION D1 appears in three rows; missing any one gives an update anomaly.

Q176 · *Application | Normalisation* MODERATE

After splitting Employee into Emp(EmpID, DeptID) and Dept(DeptID, DeptName), which problem is removed?

A. The need for any join to read employee and department data

B. Having to change DeptName in many rows (update anomaly)

C. The need for a key to identify each department

D. Repeated EmpID values in the Emp table

Ans: B

EXPLANATION DeptName is now stored once per department.

Q177 · *Scenario-based | Normalisation* DIFFICULT

Match the scenario with the anomaly: (a) a department cannot be recorded until an employee joins; (b) deleting the last employee removes the department; (c) changing a department name must be done in many rows.

A. (a) update, (b) deletion, (c) insertion

B. (a) insertion, (b) deletion, (c) update

C. (a) insertion, (b) update, (c) deletion

D. (a) deletion, (b) insertion, (c) update

Ans: B

EXPLANATION Blocked addition is insertion, unintended loss is deletion, and many-row inconsistency risk is update.

Q178 · *Concept | Normalisation* DIFFICULT

Which statement about anomalies and redundancy is correct?

A. Redundancy is itself caused by the anomalies in the tables

B. Anomalies arise only because attribute values are atomic

C. Anomalies arise when facts are repeated or mixed with unrelated facts

D. Anomalies arise only when relations are already in 3NF

Ans: C

EXPLANATION Redundant or mixed storage is the root cause; normalization separates facts to reduce it.

SECTION 09

Table and Dependency-Based Questions

20 Questions · 5 Moderate · 15 Difficult

Q179 · *Table-based | Normalisation* DIFFICULT

Table key (RollNo, SubjectCode) -> SubjectName, Marks; SubjectName depends only on SubjectCode. Which anomaly is most directly caused by this design?

A. No anomaly is possible, because Marks depends on the whole key

B. Only a deletion anomaly is possible, as update is never affected

C. Only an insertion anomaly is possible, as update is never affected

D. Update anomaly, as SubjectName repeats for each student of a subject

Ans: D

EXPLANATION Because SubjectName repeats for each student in the subject, updating it correctly everywhere risks inconsistency.

Q180 · *Decomposition | Normalisation* DIFFICULT

Same table (RollNo, SubjectCode) -> SubjectName, Marks. What decomposition removes both the 2NF violation and the associated anomaly?

A. RollNo(RollNo, SubjectName) and Subject(SubjectCode, Marks, RollNo)

B. Subject(SubjectCode, SubjectName) and Marks(RollNo, SubjectCode, Marks)

C. One single table kept unchanged with all attributes together

D. Marks(RollNo, Marks) and Subject(SubjectCode, SubjectName, RollNo)

Ans: B

EXPLANATION Isolating SubjectName with its true key (SubjectCode) removes the partial dependency and the update anomaly it causes.

Q181 · *Dependency-based | Normalisation* MODERATE

Key is EmpID. Which statement is correct?

EMPLOYEE

EmpID	EmpName	CityID	CityName
E1	Asha	C1	Mumbai
E2	Ravi	C1	Mumbai
E3	Neel	C2	Delhi

A. Non-atomic value; only an insertion anomaly can occur

B. Transitive dependency; deleting E3 loses city C2's name

C. Partial dependency; no anomaly can occur in the table at all

D. No dependency problem; the relation is fully normalized

Ans: B

EXPLANATION EmpID -> CityID -> CityName is transitive, and C2 is recorded only in E3's row.

Q182 · *Table-based | Normalisation* DIFFICULT

Table key (OrderID, ProductID) -> Quantity, ProductPrice, where ProductPrice depends only on ProductID. Choose the correct combination of violation, anomaly, and fix:

A. 1NF violation causing deletion anomaly; fixed by merging Quantity and ProductPrice

B. No violation exists; ProductPrice is correctly placed in this table

C. Transitive dependency causing insertion anomaly; fixed by deleting OrderID from the table

D. Partial dependency, update anomaly; move ProductPrice to a Product table

Ans: D

EXPLANATION ProductPrice depends on only part of the key; separating it into a Product table removes the resulting update anomaly.

Q183 · *Table-based | Normalisation* MODERATE

Which normalization step must be done first for this relation?

TEACHER

TeacherID	TeacherName	Subjects
T1	Rao	Maths, Physics
T2	Iyer	English

A. No step is needed since the table is already normalized

B. Check for partial dependency before checking atomicity

C. Check for transitive dependency before anything else

D. Convert it to 1NF with one subject in each row

Ans: D

EXPLANATION 2NF and 3NF analysis assumes 1NF; the Subjects cell holds two values.

Q184 · *Table-based | Normalisation* DIFFICULT

Table key (StudentID, CourseID) -> Grade; StudentName depends on StudentID; CourseName and CourseCredit depend on CourseID. Which normal form is satisfied, and why?

A. 1NF only; StudentName, CourseName, CourseCredit are partially dependent

B. Not even 1NF, because the key has two attributes

C. 2NF, because Grade depends on the full composite key of the table

D. 3NF, because no transitive dependency exists in the table

Ans: A

EXPLANATION All values are atomic (1NF), but multiple non-key attributes depend on only part of the composite key, so 2NF fails.

Q185 · *Table-based | Normalisation* MODERATE

Doctor D1's specialization changes to Neurology, but only P1's row is updated. What results?

PATIENT

PatientID	PatientName	DoctorID	DoctorSpecialization
P1	Anu	D1	Cardiology
P2	Bala	D1	Cardiology
P3	Chitra	D2	ENT

A. A deletion anomaly, because deleting a patient removes the doctor's details

B. An update anomaly, as specialization is repeated in each patient row

C. An insertion anomaly, because a new doctor cannot be recorded without a patient

D. No anomaly, because PatientID is unique so every row is independent

Ans: B

EXPLANATION D1 now has two different specializations in the data, which is inconsistency from redundancy.

Q186 · *Table-based | Normalisation* DIFFICULT

Table key (SupplierID, PartID) -> SupplyQuantity, SupplierCity, where SupplierCity depends only on SupplierID. What is the correct fix, and which rule does it satisfy?

A. Move SupplierCity to Supplier(SupplierID, SupplierCity) for 2NF

B. Move SupplyQuantity to a separate table so that 1NF is satisfied

C. Delete SupplierCity entirely so that 3NF is satisfied

D. No fix is needed because the relation already satisfies 2NF

Ans: A

EXPLANATION SupplierCity's dependency on only part of the composite key is a partial dependency; isolating it satisfies 2NF.

Q187 · *Table-based | Normalisation* MODERATE

Key is LoanID. Assuming 1NF holds, the highest normal form satisfied is:

LOAN

LoanID	MemberName	BookID	BookTitle	BookGenre
L1	Asha	B1	DBMS	Tech
L2	Ravi	B1	DBMS	Tech
L3	Meena	B2	Poems	Lit

A. 2NF

B. 1NF only

C. 3NF

D. None of these

Ans: A

EXPLANATION The key is single, so 2NF holds; BookTitle and BookGenre depend on BookID, a non-key attribute, so 3NF fails.

Q188 · *Dependency-based | Normalisation* DIFFICULT

Table key (ProjectID, EmployeeID) -> HoursLogged, EmployeeName, EmployeeDept, where EmployeeDept depends on EmployeeName which depends on EmployeeID. Identify all dependency issues present:

A. Only transitive dependency is present; no partial dependency exists

B. Only partial dependency is present; no transitive dependency exists

C. No issue is present at all; the relation is already in 3NF form

D. Partial (EmployeeName on EmployeeID) and transitive (EmployeeDept via EmployeeName)

Ans: D

EXPLANATION EmployeeName is partially dependent on part of the key, and EmployeeDept is transitively dependent through EmployeeName.

Q189 · *Table-based | Normalisation* MODERATE

A new owner, Kavya, who has not bought any vehicle yet, cannot be recorded in this relation (key is VehicleID). This is a:

VEHICLE

VehicleID	OwnerName	OwnerAddress
V1	Kiran	Pune
V2	Kiran	Pune
V3	Sita	Delhi

A. Update anomaly

B. Violation of 1NF

C. Deletion anomaly

D. Insertion anomaly

Ans: D

EXPLANATION Owner data can be stored only together with a vehicle, so a valid fact cannot be inserted on its own.

Q190 · *Table-based | Normalisation* DIFFICULT

Table key (StudentID, CourseID) -> Grade, CourseName; CourseName depends only on CourseID. If a course with a single enrolled student is dropped, its CourseName is permanently lost. Identify the violation, the anomaly, and the correct fix together:

A. Partial dependency causing deletion anomaly; move CourseName to a Course table

B. No violation exists; this is normal behavior and needs no fix at all

C. Transitive dependency causing an insertion anomaly; fix by deleting the Grade attribute

D. 1NF violation causing an update anomaly; fix by merging CourseName into Grade

Ans: A

EXPLANATION CourseName's partial dependency on CourseID leads to its loss upon deletion; isolating it in a Course relation prevents this.

Q191 · *Table-based | Normalisation*

DIFFICULT

Key is (StudentID, CourseID). Which description is correct?

ENROLLMENT

StudentID	CourseID	StudentName	CourseName	Instructor	Grade
S1	C1	Asha	DBMS	Rao	A
S2	C1	Ravi	DBMS	Rao	B
S1	C2	Asha	Java	Iyer	A

A. Only StudentName shows a violation of the normal form

B. StudentName, CourseName and Instructor are all partially dependent

C. Instructor shows only a transitive dependency on the key attribute

D. The relation is already in 3NF with no violation

Ans: B

EXPLANATION StudentName depends on StudentID; CourseName and Instructor depend on CourseID alone.

Q192 · *Decomposition | Normalisation*

DIFFICULT

Key is (StudentID, CourseID). Which set of relations is in 3NF?

ENROLLMENT

StudentID	CourseID	StudentName	CourseName	Grade
S1	C1	Asha	DBMS	A
S2	C1	Ravi	DBMS	B

A. Enrollment(StudentID, CourseID, StudentName, CourseName, Grade) only

B. Student(StudentID, CourseName, Grade), Course(CourseID, StudentName), Enrollment(StudentID, CourseID, Grade)

C. Student(StudentID, StudentName), Course(CourseID, CourseName), Enrollment(StudentID, CourseID, Grade)

D. Student(StudentID, StudentName, Grade) and Course(CourseID, CourseName)

Ans: C

EXPLANATION Each attribute is kept with the key it depends on completely.

Q193 · *Dependency-based | Normalisation*

DIFFICULT

Key is EmpID. Which chain is evident from the rows?

EMPLOYEE

EmpID	DeptID	DeptName
E1	D1	Sales
E2	D1	Sales
E3	D2	HR
E4	D3	IT

A. DeptName -> EmpID -> DeptID

B. EmpID -> DeptID -> DeptName

C. EmpID -> DeptName -> DeptID

D. DeptID -> EmpID -> DeptName

Ans: B

EXPLANATION Each EmpID has one DeptID, and each DeptID has one DeptName; DeptID repeats across employees.

Q194 · *Table-based | Normalisation* DIFFICULT

Key is (ProjectID, EmpID). Which statement is correct?

ASSIGNMENT

ProjectID	EmpID	Hours	ProjectManager
P1	E1	10	Rao
P1	E2	12	Rao
P2	E1	8	Iyer

A. ProjectManager is only transitively dependent on the key attributes

B. The relation is already in 3NF with no violation

C. Hours is partially dependent on ProjectID alone, so 2NF fails

D. ProjectManager is partially dependent on ProjectID; 2NF is violated

Ans: D

EXPLANATION ProjectManager repeats for every employee on the same project.

Q195 · *Table-based | Normalisation* DIFFICULT

Key is (OrderID, ItemID). Which normalization step comes first, and why?

ORDERITEM

OrderID	ItemID	ItemName	Qty
O1	I1	Pen	2
O2	I1	Pen	4
O2	I2	Ink	1

A. No step is needed, as the table is already in 3NF

B. Convert to 1NF first, because the values are not atomic

C. Remove the transitive dependency first, because 3NF is the stricter form

D. Remove the partial dependency (ItemName on ItemID), as 2NF precedes 3NF

Ans: D

EXPLANATION Normal forms are reached in order; here 1NF holds, so 2NF is next.

Q196 · *Table-based | Normalisation* DIFFICULT

Key is StudentID. After removing the transitive dependency, which pair of relations is correct?

STUDENT

StudentID	HostelID	Warden	Fee
S1	H1	Rao	3000
S2	H1	Rao	3000
S3	H2	Iyer	3500

A. Student(StudentID, Warden) and Hostel(HostelID, Fee, StudentID)

B. Student(StudentID, HostelID, Warden) and Hostel(HostelID, Fee)

C. Student(StudentID, HostelID) and Hostel(HostelID, Warden, Fee)

D. Student(StudentID) and Hostel(HostelID, Warden, Fee) with no link

Ans: C

EXPLANATION Warden and Fee depend on HostelID, so they move to a Hostel table that Student links to.

Q197 · *Dependency-based | Normalisation* DIFFICULT

Key is (SupplierID, PartID). If every row of S1 shows the same City, which anomaly is most likely when S1 moves to another city?

SUPPLY

SupplierID	PartID	Qty	City
S1	P1	10	Pune
S1	P2	5	Pune
S2	P1	7	Delhi

A. Deletion anomaly only, as supplier rows are lost

B. No anomaly, as the key is composite

C. Update anomaly, as City is repeated for each part

D. Insertion anomaly only, as no new part can be added

Ans: C

EXPLANATION City depends on SupplierID alone (partial dependency), so it is stored once per part.

Q198 · *Table-based | Normalisation* DIFFICULT

Key is EmpID. What is the highest normal form, and which rule fails next?

EMPLOYEE

EmpID	Name	Designation	Grade
E1	Asha	Manager	G1
E2	Ravi	Clerk	G3
E3	Neel	Manager	G1

A. 2NF; the 3NF rule fails because of a transitive dependency

B. 1NF; the 2NF rule fails because of a partial dependency

C. Not in 1NF; atomicity fails because a column has several values

D. 3NF; no rule fails because every attribute depends on the key

Ans: A

EXPLANATION Values are atomic and the key is single, but Grade depends on Designation.

SECTION 10

Comparison Questions

15 Questions · 3 Easy · 10 Moderate · 2 Difficult

Q199 · *Comparison | Normalisation* MODERATE

1NF vs 2NF — the key additional rule 2NF introduces over 1NF is:

A. Requirement that every table must have a composite key

B. Requirement that every attribute value must be atomic

C. Removal of transitive dependency among non-key attributes

D. Removal of partial dependency on a composite key

Ans: D

EXPLANATION 2NF's distinguishing rule, beyond 1NF's atomicity, is removing partial dependency.

Q200 · *Comparison | Normalisation* MODERATE

2NF vs 3NF — the key additional rule 3NF introduces over 2NF is:

A. Removal of transitive dependency among non-key attributes

B. A requirement that the key has a single attribute

C. Removal of partial dependency on a composite key attribute

D. A requirement that all values are atomic

Ans: A

EXPLANATION 3NF's distinguishing rule, beyond 2NF, is removing transitive dependency.

Q201 · *Comparison | Normalisation* MODERATE

1NF vs 3NF — which statement is correct?

A. 1NF includes all requirements of 3NF plus some extra rules

B. 1NF and 3NF address completely unrelated issues

C. 3NF is a weaker condition than 1NF and has fewer rules

D. 3NF includes everything required by 1NF plus further rules

Ans: D

EXPLANATION The normal forms are cumulative, so 3NF is strictly stronger than and inclusive of 1NF's requirements.

Q202 · *Comparison | Normalisation* MODERATE

Partial dependency vs transitive dependency — the key difference is:

A. Partial dependency is among non-key attributes; transitive is on part of the key

B. Partial dependency arises only in 3NF and transitive only in 2NF

C. Partial: part of a composite key; transitive: non-key depends on non-key

D. Both terms describe exactly the same kind of dependency

Ans: C

EXPLANATION This correctly distinguishes the two dependency types and the normal forms they relate to.

Q203 · *Comparison | Normalisation* EASY

Partial dependency requires:

A. A composite primary key

B. A foreign key referencing itself

C. No primary key at all

D. A single-attribute primary key

Ans: A

EXPLANATION Partial dependency, by definition, needs a multi-attribute key.

Q204 · *Comparison | Normalisation* EASY

Transitive dependency can occur:

A. Only when the primary key is composite

B. Never, once the relation is normalized

C. Only when the relation has no primary key

D. Even with a single-attribute key

Ans: D

EXPLANATION Unlike partial dependency, transitive dependency doesn't require a composite key.

Q205 · *Comparison | Normalisation* MODERATE

Insertion anomaly vs update anomaly — which best distinguishes them?

A. Insertion: unable to add valid data; update: inconsistent data after a change

B. Insertion anomaly is about losing other useful data when a row is deleted from the table

C. Update anomaly is about being unable to add valid data without other data present

D. Both anomalies describe exactly the same problem of repeated data

Ans: A

EXPLANATION These represent two distinct problems that stem from redundancy in different operational contexts.

Q206 · *Comparison | Normalisation* MODERATE

Update anomaly vs deletion anomaly — which best distinguishes them?

A. Deletion anomaly occurs only when existing values are updated in a table

B. Update anomaly risks inconsistent duplicates; deletion anomaly risks losing needed data

C. Update anomaly occurs only when new records are inserted into a table

D. Both anomalies always occur together and can never occur separately

Ans: B

EXPLANATION Update anomalies concern consistency of repeated data; deletion anomalies concern unintended data loss.

Q207 · *Comparison | Normalisation* MODERATE

Advantages vs disadvantages of normalization — which statement is accurate?

A. Normalization has only advantages and no disadvantages whatsoever

B. Normalization has only disadvantages and no advantages whatsoever

C. It cuts redundancy and anomalies but adds design and query complexity

D. Normalization removes the need for joins in every query

Ans: C

EXPLANATION Normalization involves a genuine trade-off between data quality and structural/query complexity.

Q208 · *Comparison | Normalisation* EASY

Which of these correctly pairs a normal form with the dependency it eliminates?

A. 2NF removes partial dependency; 3NF removes transitive dependency

B. 2NF eliminates transitive dependency; 3NF eliminates partial dependency

C. 3NF eliminates atomicity issues only, and 2NF eliminates nothing

D. 1NF eliminates partial dependency; 2NF eliminates non-atomic values

Ans: A

EXPLANATION This is the correct pairing of normal forms with the dependency rule each one adds.

Q209 · *Comparison | Normalisation* MODERATE

Which comparison between 1NF and 2NF is FALSE?

A. 1NF and 2NF address exactly the same rule

B. 2NF cannot be evaluated unless 1NF already holds

C. 1NF requires atomic values; 2NF additionally requires no partial dependency

D. A relation with a single-attribute key satisfies 2NF automatically once it is in 1NF

Ans: A

EXPLANATION 1NF and 2NF address different rules, so claiming they are the same rule is false.

Q210 · *Comparison | Normalisation* MODERATE

Which comparison between advantages and disadvantages of normalization is most accurate for exam purposes?

A. Advantages concern query complexity; disadvantages concern data quality and consistency

B. Advantages concern the number of tables; disadvantages concern data redundancy

C. Advantages concern data quality; disadvantages concern design and query complexity

D. Both advantages and disadvantages concern only the storage space used

Ans: C

EXPLANATION This is the standard way the trade-off is framed in database design.

Q211 · *Comparison | Normalisation* DIFFICULT

Which is true about partial and transitive dependency?

A. Both partial and transitive dependency need a composite key to occur

B. Partial needs a composite key; transitive can occur with a single key

C. Neither can occur when the key is a single attribute

D. Transitive needs a composite key; partial does not need one

Ans: B

EXPLANATION Partial dependency is defined on part of a key, so a single key cannot have one.

Q212 · *Comparison | Normalisation* DIFFICULT

A relation has a single-attribute key and is in 1NF. Compare the statements: (P) it must be in 2NF; (Q) it must be in 3NF.

A. Both are false

B. Both are true

C. P is true and Q is false

D. P is false and Q is true

Ans: C

EXPLANATION A single key rules out partial dependency but not transitive dependency.

Q213 · *Comparison | Normalisation* MODERATE

Which anomaly is about inconsistency, and which about loss of data?

A. Deletion anomaly: inconsistency; insertion anomaly: loss of data

B. Insertion anomaly: inconsistency; update anomaly: loss of data

C. Update anomaly: loss of data; deletion anomaly: inconsistency

D. Update anomaly: inconsistency; deletion anomaly: loss of data

Ans: D

EXPLANATION Repeated facts become inconsistent on update; deleting a row can remove needed facts.

SECTION 11

Difficult and Tricky Questions

17 Questions · 17 Difficult

Q214 · *Assertion-Reason | Normalisation* DIFFICULT

Assertion (A): A relation with a single-attribute primary key that is in 1NF is always in 2NF. Reason (R): Partial dependency requires a composite key.

A. Both A and R are true, but R does not explain A

B. Both A and R are true, and R correctly explains A

C. A is false, but R is true

D. A is true, but R is false

Ans: B

EXPLANATION Since partial dependency can only exist with a composite key, a single-key relation in 1NF automatically satisfies 2NF.

Q215 · *Assertion-Reason | Normalisation* DIFFICULT

Assertion (A): A relation in 3NF can never have any redundancy at all. Reason (R): 3NF eliminates every functional dependency in a database.

A. A is true, but R is false

B. A is false, but R is true

C. A is false, and R is false

D. Both A and R are true, and R explains A

Ans: C

EXPLANATION 3NF greatly reduces redundancy from partial and transitive dependencies but does not eliminate all possible redundancy or all functional dependencies, so both statements are overstated and false.

Q216 · *Statement-based | Normalisation* DIFFICULT

Statement I: Every relation in 2NF is also in 1NF. Statement II: Every relation in 1NF is also in 2NF. Which is correct?

A. Both statements are correct

B. Only Statement II is correct

C. Only Statement I is correct

D. Neither statement is correct

Ans: C

EXPLANATION 2NF is defined as 1NF plus more rules, so 2NF implies 1NF, but a 1NF relation may still have partial dependency.

Q217 · *Statement-based | Normalisation* DIFFICULT

Statement I: Transitive dependency can exist only in relations with a composite key. Statement II: Partial dependency can exist only in relations with a composite key. Which is correct?

A. Only Statement I is correct

B. Both statements are correct

C. Neither statement is correct

D. Only Statement II is correct

Ans: D

EXPLANATION Partial dependency specifically requires a composite key, but transitive dependency can occur even with a single-attribute key.

Q218 · *Case-based | Normalisation* DIFFICULT

A hospital table stores key (PatientID, TestID) -> TestResult, TestName, TestCost, where TestName and TestCost depend only on TestID. Which is the MOST complete correct diagnosis?

A. 1NF is violated, because TestResult is a non-atomic value in the table

B. No normalization issue exists, since every attribute depends on the key

C. 1NF holds but 2NF fails: TestName and TestCost depend only on TestID

D. 3NF is violated by a transitive dependency of TestCost on TestName, but 2NF is satisfied

Ans: C

EXPLANATION All values are atomic (1NF holds), but TestName and TestCost depend on only part of the composite key, violating 2NF.

Q219 · *Case-based | Normalisation* DIFFICULT

A library table: key (MemberID, BookID) -> IssueDate, BookTitle, BookAuthor, PublisherCity, where BookTitle and BookAuthor depend only on BookID, and PublisherCity depends on Publisher (tied to BookID). Which sequence of issues is present?

A. Partial dependency (BookTitle, BookAuthor) plus transitive dependency (PublisherCity)

B. No issues at all, since the design is already optimal and normalized

C. Only a transitive dependency; no partial dependency is present

D. Only a 1NF violation, due to the non-atomic IssueDate attribute

Ans: A

EXPLANATION BookTitle/BookAuthor show partial dependency, while PublisherCity is transitively tied in through Publisher — both issue types are present.

Q220 · *Tricky | Normalisation* DIFFICULT

A table has a single-attribute key and no attribute depends on another non-key attribute, but one column stores a comma-separated list of values in a single cell. What is the correct assessment?

A. The relation cannot be evaluated without a composite key

B. The relation is not even in 1NF, regardless of the key structure

C. The relation is in 3NF since there's no partial or transitive dependency

D. The relation is in 2NF but not 3NF

Ans: B

EXPLANATION A non-atomic column value violates 1NF outright, regardless of the key structure or other dependencies.

Q221 · *Tricky | Normalisation* DIFFICULT

Which of these correctly identifies a common misconception about 2NF?

A. Thinking 2NF matters only when a primary key is composite

B. Thinking 2NF and 1NF are related normal forms

C. Thinking 2NF assumes the relation is already in 1NF

D. Thinking 2NF removes transitive dependency, a 3NF rule

Ans: D

EXPLANATION A common student mistake is confusing 2NF's partial-dependency rule with 3NF's transitive-dependency rule.

Q222 · *Assertion-Reason | Normalisation* DIFFICULT

Assertion (A): A relation with a transitive dependency can still be in 2NF. Reason (R): 2NF only forbids partial dependency of non-key attributes on a composite key.

A. Both A and R are false, since 2NF also removes transitive dependency

B. A is true, but R is false because 2NF also forbids transitive dependency

C. A is false, but R is true because transitive dependency violates 2NF

D. Both A and R are true, and R correctly explains A

Ans: D

EXPLANATION 2NF does not check non-key to non-key dependency; that is the job of 3NF.

Q223 · *Assertion-Reason | Normalisation* DIFFICULT

Assertion (A): Normalization always speeds up every query. Reason (R): Normalization reduces redundancy.

A. A is false, but R is true

B. Both A and R are false

C. Both A and R are true, and R explains A

D. A is true, but R is false

Ans: A

EXPLANATION More joins can slow some queries; the reduction in redundancy is still true.

Q224 · *Statement-based | Normalisation* DIFFICULT

Statement I: A relation in 3NF is also in 2NF and 1NF. Statement II: A relation with atomic values is always in 2NF.

A. Neither is correct

B. Both are correct

C. Only Statement II is correct

D. Only Statement I is correct

Ans: D

EXPLANATION Atomic values give 1NF only; partial dependency can still exist.

Q225 · *Statement-based | Normalisation* DIFFICULT

Statement I: Removing a partial dependency may require creating new relations. Statement II: Removing a transitive dependency never needs new relations.

A. Both are correct

B. Only Statement I is correct

C. Only Statement II is correct

D. Neither is correct

Ans: B

EXPLANATION Both normal forms are reached by decomposition, i.e. new relations.

Q226 · *Case-based | Normalisation*

DIFFICULT

Key is (CourseID, StudentID). Marks depends on both. Instructor depends on CourseID. InstructorRoom depends on Instructor. Which set of issues is present?

A. No issues, since the relation is already in 3NF and 2NF

B. Partial (Instructor on CourseID) and transitive (InstructorRoom via Instructor)

C. Only a partial dependency, since InstructorRoom depends on CourseID

D. Only a transitive dependency, since Marks depends on the full key

Ans: B

EXPLANATION Instructor depends on part of the key, and InstructorRoom depends on a non-key attribute.

Q227 · *Table-based | Normalisation*

DIFFICULT

Key is (StudentID, CourseID). Which is the highest normal form, and what is the first change needed?

RESULT

StudentID	CourseID	CourseName	Marks
S1	C1	DBMS	80
S2	C1	DBMS	70
S1	C2	Java	65

A. 1NF; remove partial dependency of CourseName on CourseID

B. 2NF; remove the transitive dependency from the relation first

C. Not in 1NF; split the non-atomic values first

D. 3NF; no change is needed in the relation

Ans: A

EXPLANATION All values are atomic, but CourseName depends on CourseID alone.

Q228 · *Tricky | Normalisation*

DIFFICULT

A relation has the key StudentID and the attributes Name and Age only. Each value is atomic. Which statement is true?

A. It is not in 3NF because Age may repeat

B. It is in 2NF only

C. It is in 1NF, 2NF and 3NF

D. It is in 1NF only

Ans: C

EXPLANATION Repeating values such as the same Age are not a dependency problem; no partial or transitive dependency exists.

Q229 · *Tricky | Normalisation* DIFFICULT

Two employees in an Employee table (key EmpID) have the same Salary. Does this make the relation violate 3NF?

A. Yes, because repeated Salary values violate 1NF atomicity

B. Yes, because Salary values are repeated, so the relation contains redundancy

C. No, equal values do not make one non-key attribute depend on another

D. Yes, because repeated Salary creates a partial dependency on EmpID

Ans: C

EXPLANATION A transitive dependency needs one non-key attribute to determine another; equal values are not enough.

Q230 · *Tricky | Normalisation* DIFFICULT

In which case can a partial dependency NOT exist?

A. When the primary key has two attributes together

B. When the primary key has a single attribute

C. When the relation is already in 1NF but has a composite key

D. When the relation has a foreign key referring to another table

Ans: B

EXPLANATION Partial means dependence on part of a key, so a one-attribute key has no 'part'.

SECTION 12

Mixed OAVS TGT Practice Set

19 Questions · 4 Easy · 11 Moderate · 4 Difficult

Q231 · *Fill-in-the-Blank | Normalisation* EASY

1NF requires that all attribute values be _____.

A. Transitive
B. Atomic
C. Composite
D. Redundant

Ans: B

EXPLANATION Atomicity (indivisibility of values) is the core requirement of 1NF.

Q232 · *Fill-in-the-Blank | Normalisation* EASY

The dependency of a non-key attribute on only part of a composite key is called _____ dependency.

A. Full
B. Transitive
C. Multivalued
D. Partial

Ans: D

EXPLANATION This is, by definition, a partial dependency, which 2NF eliminates.

Q233 · *Fill-in-the-Blank | Normalisation* EASY

The dependency of a non-key attribute on another non-key attribute is called _____ dependency.

A. Composite
B. Transitive
C. Atomic
D. Partial

Ans: B

EXPLANATION This describes transitive dependency, which 3NF eliminates.

Q234 · *Fill-in-the-Blank | Normalisation* MODERATE

A table where deleting a row unintentionally removes needed information suffers from a _____ anomaly.

A. Structural
B. Insertion
C. Deletion
D. Update

Ans: C

EXPLANATION Unintended loss of information upon deletion defines a deletion anomaly.

Q235 · *Match-the-Following | Normalisation* MODERATE

Match: (1) 1NF, (2) 2NF, (3) 3NF with (a) removes transitive dependency, (b) removes non-atomic values, (c) removes partial dependency. Correct matching:

A. 1-a, 2-b, 3-c

B. 1-c, 2-a, 3-b

C. 1-b, 2-a, 3-c

D. 1-b, 2-c, 3-a

Ans: D

EXPLANATION 1NF removes non-atomic values (b), 2NF removes partial dependency (c), 3NF removes transitive dependency (a).

Q236 · *Match-the-Following | Normalisation* MODERATE

Match: (1) Insertion anomaly, (2) Update anomaly, (3) Deletion anomaly with (a) inconsistent duplicate data, (b) cannot add valid data, (c) unintended loss of data. Correct matching:

A. 1-b, 2-c, 3-a

B. 1-a, 2-b, 3-c

C. 1-b, 2-a, 3-c

D. 1-c, 2-b, 3-a

Ans: C

EXPLANATION Insertion anomaly (b) blocks valid insertion, update anomaly (a) causes inconsistency, deletion anomaly (c) causes data loss.

Q237 · *Statement-based | Normalisation* MODERATE

Which statement is TRUE regarding normalization and anomalies?

A. Normal forms have no relationship with anomalies whatsoever

B. Higher normal forms generally lower the risk of anomalies

C. Higher normal forms always raise the risk of anomalies

D. Only 1NF has any effect on the risk of anomalies

Ans: B

EXPLANATION As redundancy decreases with higher normal forms, the risk of anomalies also decreases.

Q238 · *Case-based | Normalisation* MODERATE

A school database stores key (TeacherID, ClassID) -> Subject, TeacherPhone, where TeacherPhone depends only on TeacherID. Which normalization step should be applied first?

A. Merge ClassID into TeacherPhone to form a single combined attribute

B. Delete TeacherID from the table to avoid any dependency on it

C. Move TeacherPhone to a separate Teacher table (removes partial dependency)

D. Convert Subject into multiple columns, one for every class taught by the teacher

Ans: C

EXPLANATION TeacherPhone's partial dependency on TeacherID should be resolved by isolating it in a Teacher table.

Q239 · *Application | Normalisation* MODERATE

An unnormalized 'Sales' table repeats CustomerAddress in every sale record. After normalization, CustomerAddress should be stored:

A. Once in a Customer table, referenced by CustomerID in Sales

B. Repeated in every Sales row exactly as it was before

C. Stored only inside the primary key of the Sales table

D. Removed from the database entirely, so no address is kept

Ans: A

EXPLANATION This is the standard normalization fix — isolating repeated attributes into their own table linked by a key.

Q240 · *Scenario-based | Normalisation* MODERATE

A relation key (SubjectID, TeacherID) -> ClassroomNo, TeacherDept, where TeacherDept depends only on TeacherID. Which normal form must be targeted to fix this, and what is removed?

A. 3NF; transitive dependency of TeacherDept on the key

B. None; the relation is already in 2NF

C. 2NF; partial dependency of TeacherDept on TeacherID

D. 1NF; non-atomic values of ClassroomNo

Ans: C

EXPLANATION TeacherDept depends on only part of the composite key, corrected by reaching 2NF.

Q241 · *Error-identification | Normalisation* MODERATE

Identify the error in this claim: 'A relation in 2NF must automatically be in 3NF since 3NF is just a stricter version of 2NF.'

A. Incorrect; a 2NF relation may still have a transitive dependency

B. The claim is correct; 2NF already removes every transitive dependency

C. The claim is incorrect because 2NF is stricter than 3NF

D. The claim is correct because 3NF is a weaker version of 2NF

Ans: A

EXPLANATION Being stricter does not mean automatic — a 2NF relation can still contain a transitive dependency, failing 3NF.

Q242 · *Error-identification | Normalisation* DIFFICULT

Identify the error: 'Normalization always improves query performance because it removes redundant data.'

A. Irrelevant, since normalization has nothing to do with performance

B. Fully correct, with no exceptions at all for any query

C. Correct only for 1NF, but not for 2NF or 3NF designs

D. Incorrect; normalization adds joins and query complexity, which can slow some queries

Ans: D

EXPLANATION While normalization improves data quality, it can add join overhead, a genuine disadvantage/trade-off.

Q243 · *Case-based | Normalisation* DIFFICULT

A relation key (OrderID, ProductID) -> Quantity, CustomerName, ProductCategory, CategoryDiscount, where CustomerName depends only on OrderID, ProductCategory depends only on ProductID, and CategoryDiscount depends on ProductCategory. List all issues present.

A. Partial (CustomerName, ProductCategory) and transitive (CategoryDiscount)

B. Only a 1NF violation, because Quantity is non-atomic

C. No issues exist; the relation is already in 3NF

D. Only a transitive dependency exists; no partial dependency is present

Ans: A

EXPLANATION CustomerName and ProductCategory each show partial dependency, and CategoryDiscount shows transitive dependency via ProductCategory.

Q244 · *Application | Normalisation* MODERATE

For the OAVS TGT exam, which single principle best summarizes when to stop normalizing a simple relation for practical purposes?

A. Stop once the relation is in 3NF, with no partial or transitive dependency

B. Stop only once every attribute has become a primary key of its own table

C. Never stop; keep splitting the tables until every table has only a single column

D. Stop as soon as the relation has been split into more than one table

Ans: A

EXPLANATION For most practical and exam purposes within this syllabus, 3NF — free of partial and transitive dependency — is the target stopping point. Page 54 of 54

Q245 · *Match-the-Following | Normalisation* MODERATE

Match the problem with its normal form: (1) multi-valued cell, (2) partial dependency, (3) transitive dependency, with (a) 3NF, (b) 1NF, (c) 2NF.

A. 1-a, 2-c, 3-b

B. 1-b, 2-c, 3-a

C. 1-b, 2-a, 3-c

D. 1-c, 2-b, 3-a

Ans: B

EXPLANATION Atomicity relates to 1NF, partial dependency to 2NF and transitive dependency to 3NF.

Q246 · *Fill-in-the-Blank | Normalisation* EASY

A relation in 3NF is always also in _____ and 1NF.

A. 2NF

B. Atomic form only

C. Partial form

D. Transitive form

Ans: A

EXPLANATION The normal forms are cumulative: 3NF includes 2NF, which includes 1NF.

Q247 · *Fill-in-the-Blank | Normalisation* MODERATE

Being unable to add a department until an employee exists is an _____ anomaly.

A. deletion

B. atomic

C. insertion

D. update

Ans: C

EXPLANATION A valid fact cannot be inserted on its own.

Q248 · *Table-based | Normalisation* DIFFICULT

Which statement about this relation (key is (StudentID, CourseID)) is correct?

ENROLLMENT

StudentID	CourseID	CourseName	Grade
S1	C1	DBMS	A
S2	C1	DBMS	B
S3	C1	DBMS	A

A. It violates 3NF only, and the deletion anomaly appears when a row is removed

B. It is in 3NF, and no anomaly is possible when names change

C. It violates 2NF; update anomaly appears when a course name changes

D. It violates 1NF, and the insertion anomaly appears when a student is added

Ans: C

EXPLANATION CourseName depends on CourseID alone; changing it means editing several rows.

Q249 · *Case-based | Normalisation* DIFFICULT

A school keeps Teacher(TeacherID, Name, Subjects) with Subjects such as 'Maths, Physics'. List the correct order of steps to reach a clean design.

A. Check partial dependency first, then atomicity, and then 3NF

B. Check transitive dependency first, then 2NF, and then 1NF

C. Only 3NF needs to be checked, as 1NF and 2NF follow automatically

D. Make Subjects atomic (1NF), then check 2NF, then check 3NF

Ans: D

EXPLANATION Normal forms are reached in order: 1NF, 2NF, 3NF.